

EXCISE AND TAXATION MANAGEMENT SYSTEM FOR BALTISTAN



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CERTIFICATE

This is certified that we have read and checked the work contained in the final year project report as confirmation of the standard required for partial fulfillment of the degree of Bachelor of Science in Computer Science (BSCS) session 2020-2024. It is our judgment that the final year project report is up to the standard to warrant its acceptance to the University of Baltistan, Skardu (UOBS) for the degree of Bachelor of Science in Computer Science.

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Declaration

We hereby declare that this undergraduate project, neither as a whole nor as a part there of has been copied out from any sources, wherever references have been provided. It is further declared that we have developed this undergraduate project and accompanied report entirely on the bases of our knowledge. If any part of this undergraduate project is proved to be copied out from any source and found to be reproduced of some other source without proper acknowledgement, we will stand by the consequences, and the university will take no responsibility.

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Abstract

Our team has undertaken the development of the project aimed at enhancing the management processes within the Excise and Taxation domain. This initiative primarily focuses on streamlining vehicle-related operations especially the non-custom paid ones, catering to the requirements of database management, driver verification, and public awareness regarding narcotics.

Comprised of four members, our collaborative effort has resulted in the creation of a multifaceted Management System. The system encompasses six core domains: Database Management Systems tailored for Non-Custom Paid (NCP) vehicles, Database Management for Regular / Normal Vehicles, Database Management for Rent-A-Car/Transport companies, Verification of drivers' license, Narcotics awareness and helpline portal and Fare details portal. These systems are designed to facilitate data organization and retrieval, ensuring operational easiness.

We have also integrated a driver's licensing verification for our project to improve road safety and enforcement level. Moreover, it has its own Narcotics License and Complain Portal. In addition, a Fare Details Portal has been introduced to provide clear details of fares for different destinations to and from Skardu, including the freight charges.

The goal of our project is to make further ease and efficiency in the area. We hope our all-around solution (current project) will go a long way in helping these areas offer better governance and public service to their citizens.

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Chapter 1

Introduction

1.1. Introduction

One of the most important organizations, which have to comply with regulations of vehicles is the Excise and Taxation Department a governmental body responsible for vehicle registration and token collection. In Gilgit Baltistan, they also handles the registration and regulation of Non-Custom-Paid vehicles also. They require a simple yet effective system to handle all of these processes and enhanced delivery to the public.

The project focuses on building a user interface enabling vehicles registration and regulation management for the Excise & Taxation Department. In particular, we want to build a solution that can manage most types of vehicles, both owned, irregular taxes which does not have proper tax documents or Non-Custom-Paid (NCP vehicle) and company-rented cars. On the same note, our system will also contain characteristics like searching if a driver is licensed or not, displaying drug info for awareness and performance of adding fare details for transportation services. The functionality will enable the department to streamline processes and enhance public service through providing these features. Our project seeks to harness technology to address the challenges faced by the Excise and Taxation Department, ultimately contributing to more efficient operations and better service delivery for the public.

1.2. Purpose of the Project

Project has been initiated because it is felt that the Excise and Taxation Department Baltistan needed to be modernized/restyled so as to meet with present day requirements. Working together with stakeholders and reviewing the existing systems, we realized that a major piece of puzzle is missing in terms to efficiently (in time wise - as current takes hell lot of time) manage vehicle related processes & guarantee compliance on this front. The stakeholders were aware of the scenario which was being faced by them. The visionary Director had idea of what they are facing and what they need, in order to maintain the record in less time than before. He was in search of developers who can design and develop the required system for his department.

Moreover, the decision to embark on this project stems from observing the streamlined systems in other provinces of Pakistan, such as Punjab [1], KPK [2] and Sindh [3], it became evident that Baltistan could benefit significantly from a similar approach. The

motivation behind this project is to introduce a centralized and automated system that enhances efficiency, transparency, and trust within the department, aligning with global trends towards digitization.

Our project is meant to transform the current System by developing holistic Management Solution that fits for the Excise and Taxation Department in full. The system is meant to streamline a number of vehicle management functions including title, monitoring and compliance. Expanding public safety with driver license verification mechanisms is another target of this project.

1.2.1. Objectives of the Project:

- Develop a centralized database management system for NCP vehicles, rent-a-car services, and transport companies to streamline data management and accessibility.
- Create a system for automated renewal notifications to enable permits and certificates are renewed on time, minimizing non-compliance notices and penalties.
- Promote accountability to increase transparency and trust, by providing meaningful information for stakeholders.
- Enable informed decision-making by offering rich analytics and valuable insights to the concerned departments.
- Align to the trend of digitization at a global level by bringing in systems and technology that are contemporary.
- Have strong safety features in place to protect sensitive data and prevent fraud.

While working towards these goals the project also takes a vision of helping resolve both immediate problems being faced by Excise and Taxation department today, while fitting into bigger picture objectives like improving governance/public safety; and service delivery. We are focused on delivering a solid solution that will meet the needs of the Department and we plan to do this through taking our time with planning, before executing.

1.3. Scope of the Project

Baltistan Excise & Taxation Management System is a complete solution that meets all the demands of an excise department for registration and taxation in Baltistan. The project, meant to resolve issues of the Excise and Taxation Department along with a possibility for annual token taxes collection.

The project consists of creating a centralized platform for better data management, automation and transparency across the department. The system is designed to provide a robust administrative tool and enhanced service delivery to the user by using data sources, as well as ease compliance requirements.

This will be done to make sure that the system is scalable, so as it grows in future. Security efforts are also increased. So, with each iteration growth of sensitive information, we need security around access not authorized parties full confidence. In addition, the project is to highlight user-centric design and access mechanisms so that organizational staffs who are neither formally trained nor proficient in technology make quick wins by using navigational features of audience most important, minimal training required.

Not to mention, the ability for token taxes provides governments with a strong indication of its use case as an essential tool in raising government revenue and expanding service provision. In the grander scheme of things, The Excise and Taxation Management System for Baltistan is a small step of modernizing governmental operations thus efficiency in vehicle-related processes can be upheld that benefits both department as well as public it serves.

1.4. Tools and Technologies Required

This project will use a variety of well-established technologies to provide a robust, user-friendly, and secure vehicle management system for the Excise and Taxation Department.

1.4.1. Programming Language:

Along with HTML [4] and CSS [5], we have selected PHP [6] as the primary backend programming language for this project. We also used jQuery [7] and Bootstrap [8] to improve user experience. We have chosen PHP for several reasons.

- PHP excels in processing user input and interacting with databases on the server, making it ideal for handling automobile registration processes, data manipulation, and creating dynamic web pages.
- PHP is a free and open-source language with a strong community and several resources available. This enables easier development, troubleshooting, and integration with existing systems.
- PHP provides a strong foundation for secure development methods. By following these guidelines and using prepared statements for database interaction, we can reduce vulnerabilities like as SQL injection attacks.

1.4.2. Development Environment:

We have used Visual Studio Code [9] for the development process. Visual Studio Code provides capabilities such as code completion, syntax highlighting, debugging tools, and version control integration, which considerably improve development efficiency.

1.4.3. Database:

MySQL [10] is a widely used, open-source relational database management system (RDBMS) that is well-suited for this project. The reasons behind our selection of MySQL as the Database Management System are explain below:

- **Structured data storage:** MySQL excels at storing and managing the structured data required for vehicle registration, including vehicle information, owner details, permits, and tax records.
- **Scalability:** MySQL can efficiently handle large datasets and is scalable to accommodate future growth in the Excise department data volume.
- **Integration with PHP:** MySQL offers seamless integration with PHP, allowing for efficient data manipulation and retrieval within the web application.

1.4.4. Operating Systems:

The web application was developed to work seamlessly on a variety of operating systems, including Windows, macOS, and Linux. This ensures that excise and taxation Baltistan's department staff can access information regardless of the operating system they choose.

1.4.5. Network Protocol:

The project is web based therefore it will use HTTP, the standard protocol for communicating with web servers and browsers. This means that the system can be

accessed using any modern web browser. For a more secure version we will use HTTPS protocol in order to communicate with the web server in a more secure channel.

1.5. Project Plan

Table 1: Project Plan

S. No	Milestone	Description	Expected Completions Date
1	Requirement gathering	Acquiring all the requirements from Excise and Taxation Department	20/10/2023
2	Verification of driver license	A portal where the licenses of users can be verified	30/10/2023
3	Fare details portal	A portal to provide Fare and Freight details	31/12/2023
4	Database management of NCP vehicles	A whole database of the NCP vehicles	31/01/2024
5	Database management of normal (Regular) vehicles	A whole database of the Normal vehicles	28/02/2024 (Not Required)
6	Narcotics license and complain portal	A portal to generate tobacco license	25/03/2024
7	Database management of Rent A Car / Transport companies	A whole database of the travelling companies and car rental companies	10/05/2024
8	Testing	It includes testing of whole project	15/05/2024
9	Revision	All required changes	30/06/2024
10	Deployment	Final deployment of the project	10/07/2024

1.6. Problem Statement

In Baltistan, we are dealing with a lot of paperwork and outdated ways of managing vehicles, car rentals and transport companies. This causes problems like missed renewals, financial losses, and trust issues. We are working on a solution that is easy to use and helps everyone involved.

Chapter 2

Literature Review

2.1. Background

The decision to undertake this project stems from a recognition of the pressing need to modernize and streamline the operations of the Excise and Taxation Department. Through consultation with stakeholders and an analysis of existing systems, it became evident that there was a significant gap in the department's ability to efficiently (in terms of time, because, current system is time consuming) manage vehicle-related processes and ensure regulatory compliance. The stakeholder was aware of the scenario which was being faced by them. The visionary Director had idea of what they are facing and what they need, in order to maintain the record in less time than before. He was in search of developers who can design and develop the required system for his department.

Moreover, the decision to embark on this project stems from observing the streamlined systems in other provinces of Pakistan, such as Punjab, KPK and Sindh, it became evident that Baltistan could benefit significantly from a similar approach. The motivation behind this project is to introduce a centralized and automated system that enhances efficiency, transparency, and trust within the department, aligning with global trends towards digitization.

2.2. Related Work / Literature Review

If we talk about other provinces of Pakistan, such as Punjab, KPK, and Sindh, they have their own Excise and Taxation system and people can easily find their vehicle details with a single click. Therefore, we should implement this system in the Baltistan region so that people can easily apply for vehicle verification, check the number plates of vehicles, and find their details, such as ownership, permits and other vehicle details.

This system can solve many problems of our society, such as reducing the cost of the manual system and increasing tax collection. It is also helpful for the Excise department in managing the database and handling the department's documentation more easily.

It is to be noted that they have a management system for Regular / Normal Vehicles, but the problem is that their management system is very hard to operate. As they have no technical employees, so they were facing difficulties in operating and maintaining that management system. The main fault of their existing system was that the system does

everything separately instead of doing everything in a single loop. So, they were in dire need of a user-friendly management system.

2.3. Comparative Study

Basically, compared to the management system of normal vehicle at present, our project has made great progress both in usability and functionality. The existing system is hard to operate because of the features and functionalities provided in an old architecture which also does not have a technical support, our proposed project will address these problems by providing easy operations interface as well all controls.

The major flaws in current system are the piecemeal manner of functions, where each activity to be performed independently instead as part and parcel of a unified workflow. On the contrary, our project markets an integrated one-stop solution that simultaneously simplifies processes by integrating many features into a single user-friendly product. This ensures that operational efficiency is offered and provides for lower complexities to manage the same as with any other type of vehicle.

In addition, User experience was implemented to our project in general from an intuitive interface with a "zero-knowledge" approach to the inclusion of non-tech savvy users to interact with the system with the help of better user experience. This will help the department in solving a challenge it has while running and maintaining the current system without any support from technical staff. Our goal move with the project is to let users manage a regular car, without needing an extensive technical knowledge by simplifying repetitive tasks and giving them clear instructions on how it works.

Additionally, we put a lot of focus on user experience in our project, so the UX is also very nice with intuitive interfaces and humans-friendly features. This was also in response to a problem that the department had been facing when trying to run and upkeep their existing system without any technical staff. We're empowering users to easily control all the entities (NCP vehicles, Rent A Car/Transport companies, Normal(regular) Vehicles, drivers' liscence verification etc. and tasks in streamlined way that will be hassle free for user through making task simple/reducing redundancy of instruction on each point). Through its integrated approach and emphasis on usability, our project represents a

significant advancement over existing systems, providing a comprehensive and user-friendly solution for the Excise and Taxation Department of Baltistan.

Chapter 3

System Design

3.1. Introduction

In this chapter, we lay down the whole architecture of our project. Just like a blueprint guides construction, our design sets the path for how our project will function. We will explore the basic practical detail of our project's architecture.

As we know that communication is the key to understand anything. Therefore, we make sure everyone in our project group member and other stakeholder understands our design so that we can all work together smoothly.

As we dive deeper into this chapter, we will uncover the secrets of System Design and learn how the project's anatomy is crafted with clarity and precision.

3.2. Software Architecture

When designing the architecture for the Excise and Taxation Management System for Baltistan, several meetings both online and in-person organized to ensure the system's effectiveness, maintainability, and scalability. These decisions revolve around determining the application type, deployment strategy, appropriate technologies, quality attributes, and crosscutting concerns.

3.2.1. Application Type

For the Excise and Taxation Management System, which serves as a centralized platform for data management and service delivery, a web-based application architecture is most suitable. This approach allows for easy access from various devices and locations, facilitating widespread adoption and usability among departmental staff and stakeholders. However, to check route permits, driver's licenses

3.2.2. Deployment Strategy

Choosing the right deployment strategy is essential for ensuring efficient distribution and management of the system. In the case of the Excise and Taxation Management System, a cloud-based deployment strategy offers numerous advantages, including scalability, flexibility, and cost-effectiveness. By leveraging cloud infrastructure and services, the system can accommodate fluctuating demand, scale resources as needed, and minimize upfront infrastructure costs.

3.2.3. Appropriate Technologies:

Selecting appropriate technologies is critical for building a robust and functional system. For the Excise and Taxation Management System and a combination of proven technologies such as PHP for backend development, HTML / CSS / JavaScript for frontend development, and MySQL for database management would be suitable. Additionally, by leveraging modern web development tools and libraries such as jQuery and Bootstrap we ensured compatibility with evolving standards and technologies.

3.2.4. Quality Attributes:

Identifying and prioritizing quality attributes such as performance, reliability, security, and scalability are essential for designing a successful architecture. For the Excise and Taxation Management System, emphasis should be placed on security measures to protect sensitive data, performance optimizations to ensure responsiveness under varying loads, and scalability to accommodate future growth and user demands.

3.2.5. Crosscutting Concerns:

Addressing crosscutting concerns such as logging, authentication, authorization, and error handling is crucial for ensuring consistency and maintainability across the system. By implementing reusable components and design patterns to handle these concerns, the Excise and Taxation Management System can maintain a cohesive architecture while reducing redundancy and complexity.

By carefully considering these key decisions and principles, the architecture of the Excise and Taxation Management System for Baltistan can be effectively designed to meet the department's requirements, promote usability and extendibility, and minimize costs and maintenance requirements.

3.2.6. Architectural View of whole system

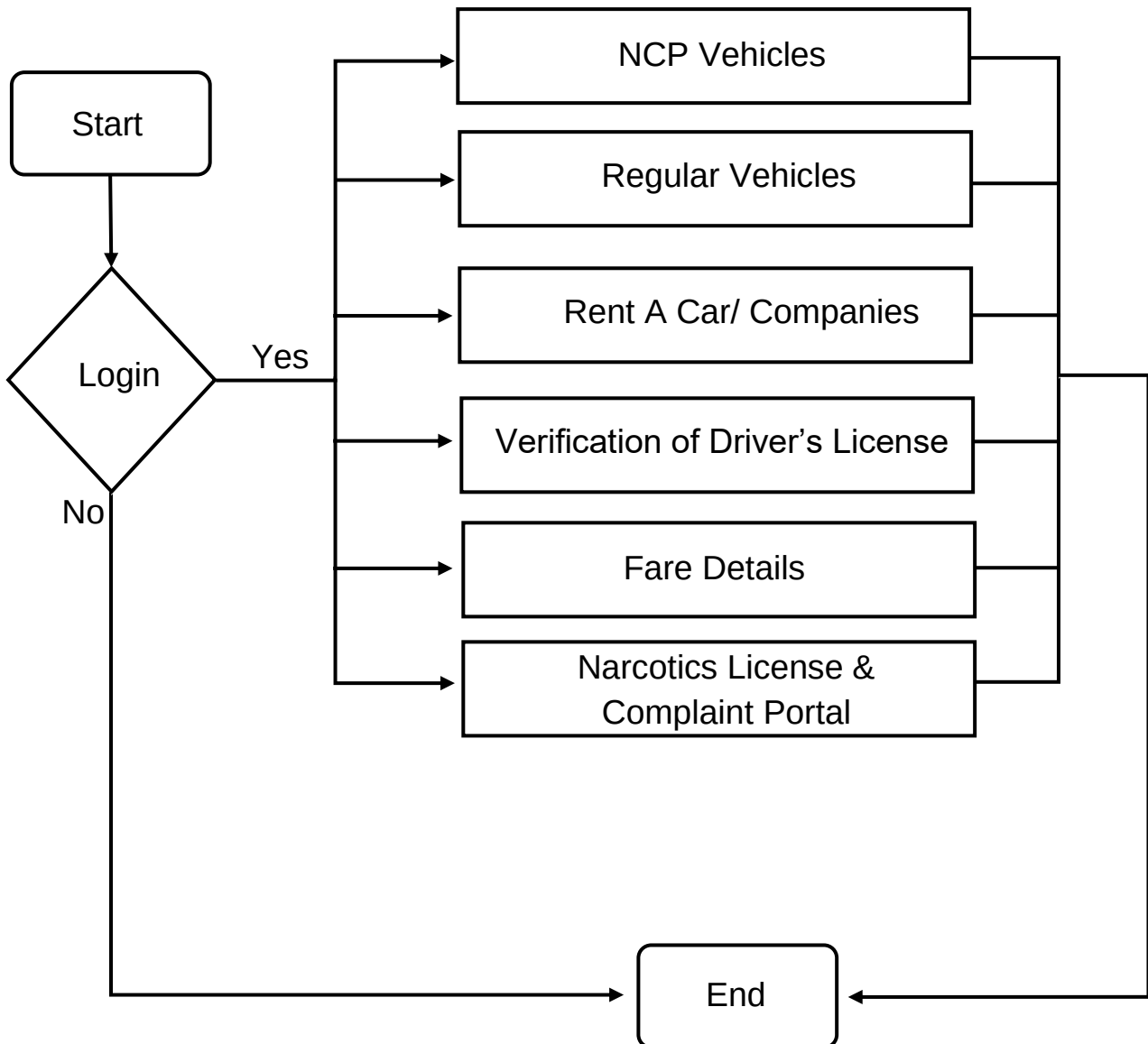


Figure 1: Whole System Architecture

Figure 1 illustrates the design of whole project. It includes the following modules:

- i. Database management for NCP vehicles.
- ii. Database management for rent a car / transport companies.
- iii. Database management for normal (regular) vehicles.
- iv. Verification of driver license.
- v. Narcotics license and complain portal.
- vi. Fare details portal.

3.2.6.1. Architecture of NCP Vehicles Registration Process

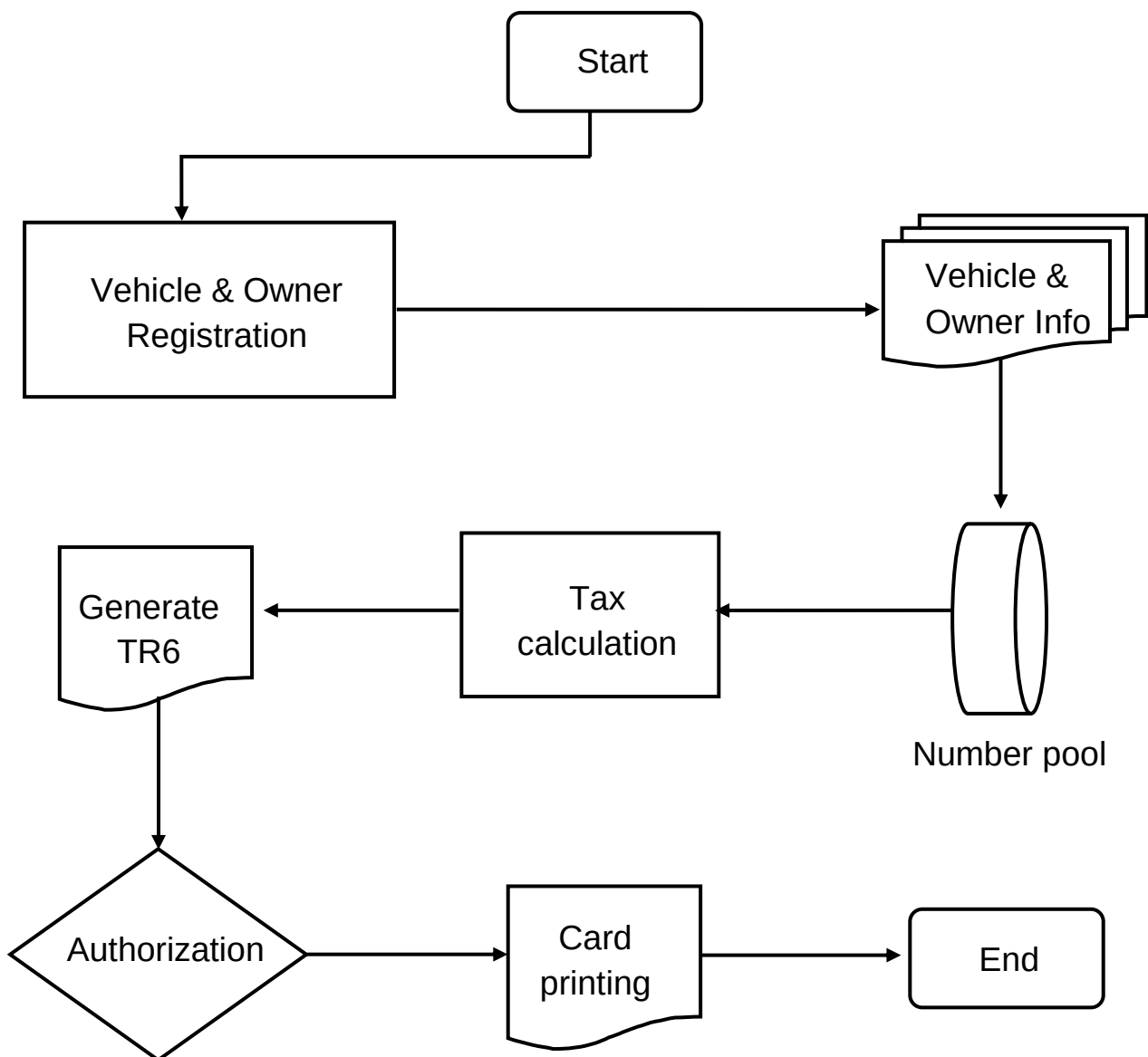


Figure 2: Architecture of NCP Vehicles Registration Process

Figure 2 illustrates the design of Database management for NCP vehicles. First of all, a form will be there, acquiring information about the vehicle which is being registered. After that, a number from existing number pool is assigned to the vehicle and the process moves towards calculating token tax according to model and CC. A challan is the printed and handed over the customer. The user will then authorize him after verification of paid challan.

3.2.6.2. Architectural View of Database Management for Rent A Car / Transport Companies

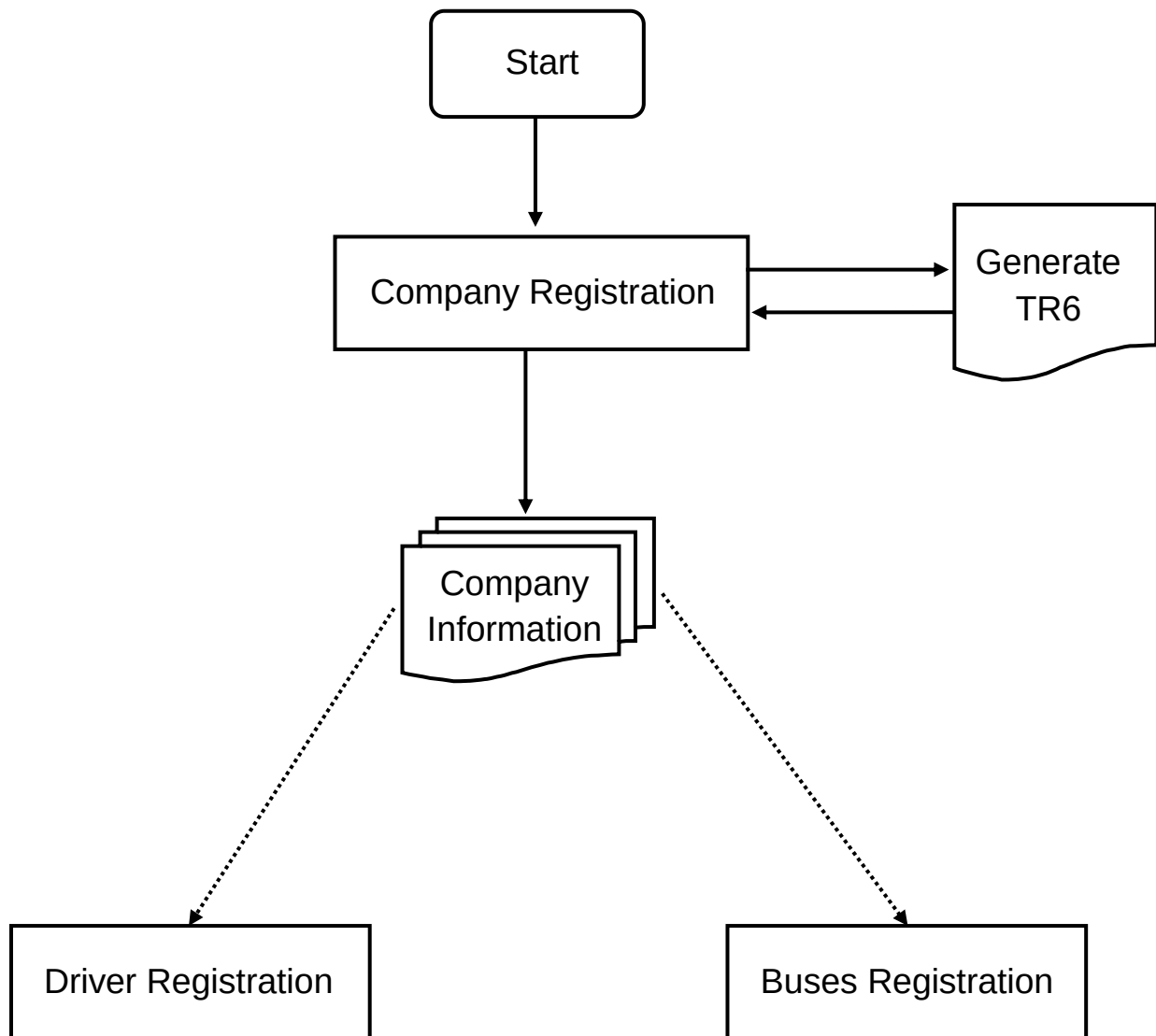


Figure 3: Architecture of Database Management for Rent A Car / Transport Companies

Figure 3 illustrates the design of database management for Rent A Car / Transport companies. First of all, transport company will be registered. After that drivers and vehicles (buses) will be added against registered companies.

3.2.6.3. Architectural View of Database Management for Normal (Regular) Vehicles.

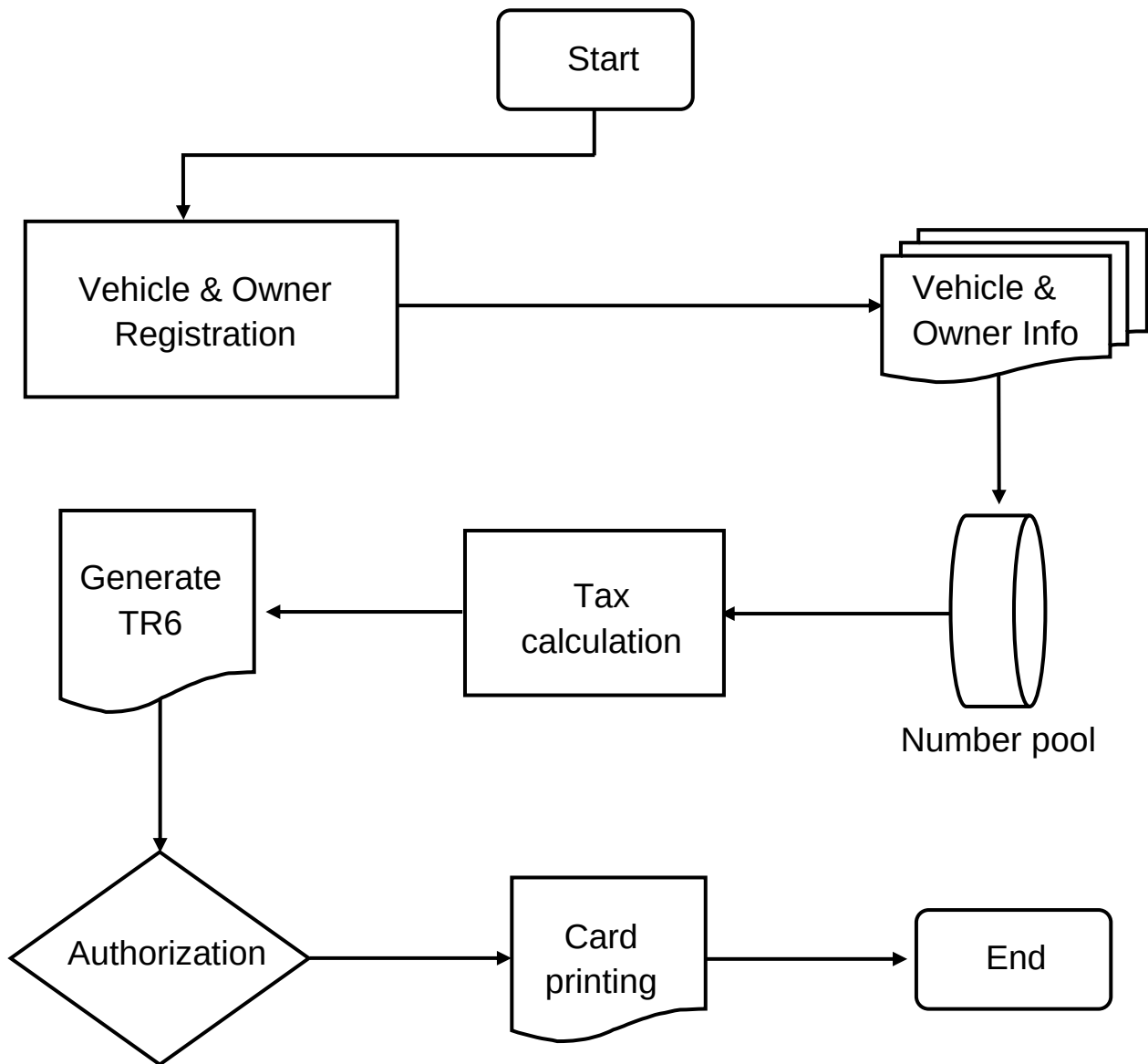


Figure 4: Architecture of Database Management for Normal (Regular) Vehicles Module

Figure 4 illustrates the design of Database management for normal vehicles. First of all, a form will be there, acquiring information about the vehicle which is being registered. After that, a number from existing number pool is assigned to the vehicle and the process moves towards calculating token tax according to model and CC. A challan is then printed and handed over the customer. The customer will pay the challan and will bring it back. The user will then authorize him.

3.2.6.4. Architectural View of Verification of Driver License Verification

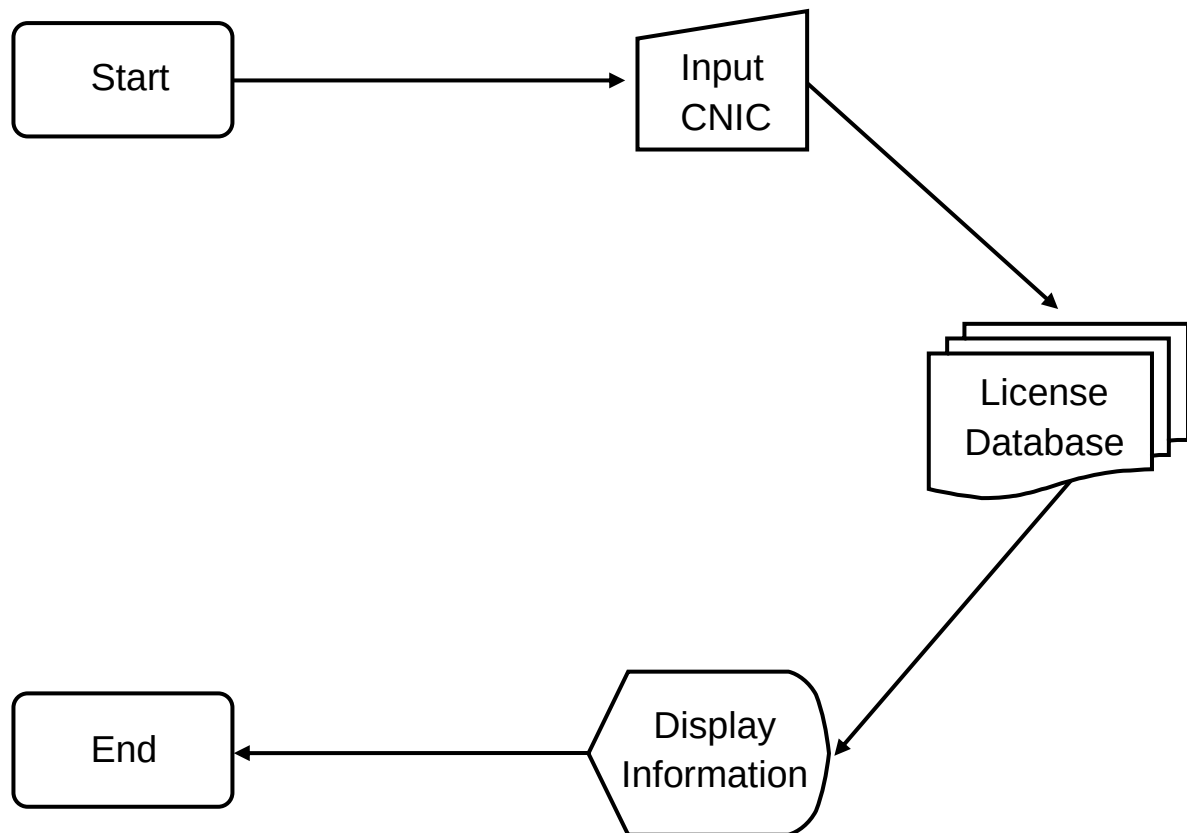


Figure 5: Architecture of Driver License Verification Module

Figure 5 illustrates the design of Driver's License Verification Module. First of all, CNIC will be fed to the system, it will retrieve details of driver if data already exists.

3.2.6.5. Architectural View of Narcotics License and Complain Portal.

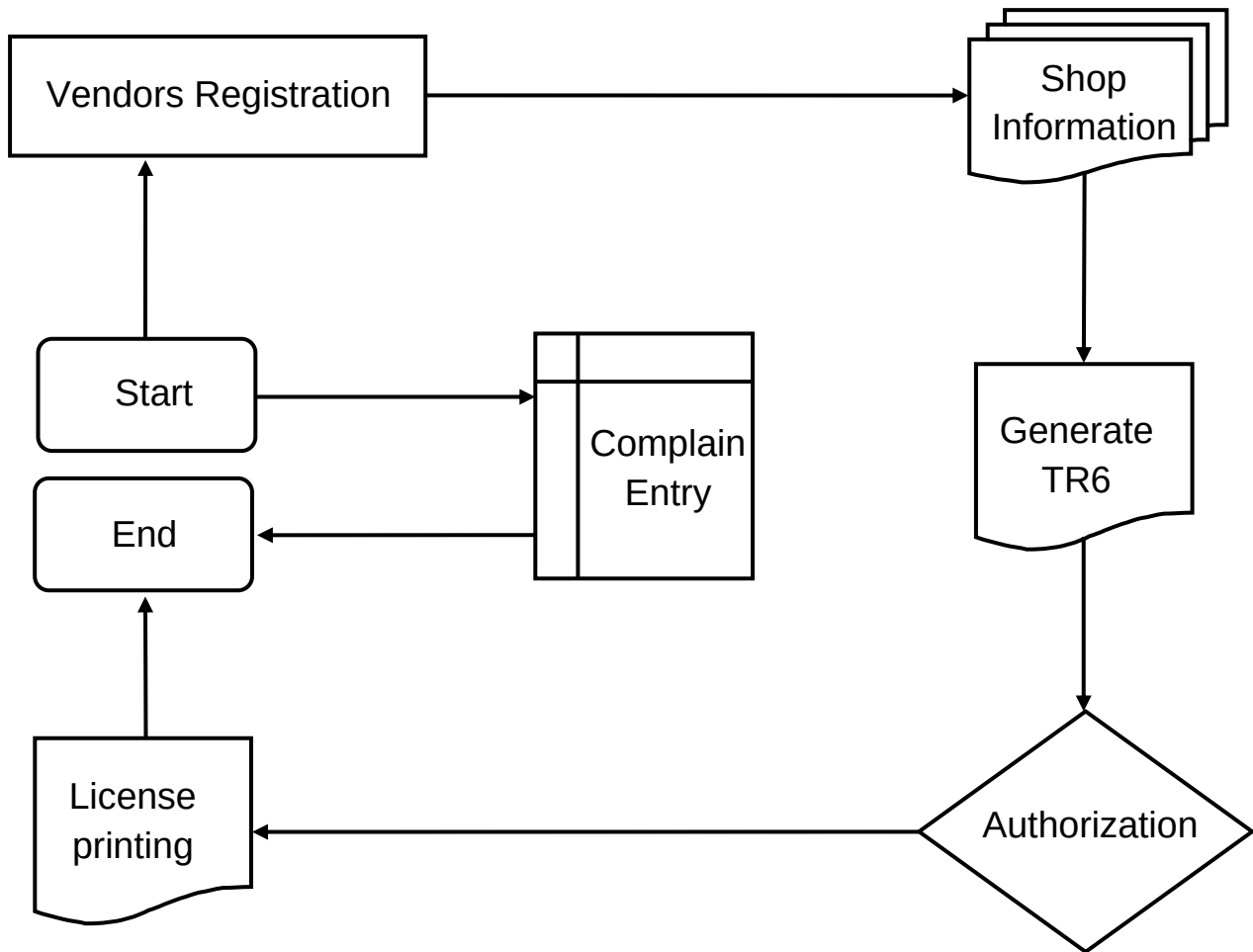


Figure 6: Architecture of Narcotics License Module

Figure 6 indicates the design and architecture of Narcotics license module. A vendor or a shopkeeper must be registered before the registration of shop. A shop is the registered against the vendor and then TR6 is generated (commonly referred as challan). After the authorization of the paid challan, a license is then printed and handed over the vendor. It is to be noted that complains can also be registered regarding narcotics related issues.

3.2.6.6. Architectural View of Fare Details Portal.

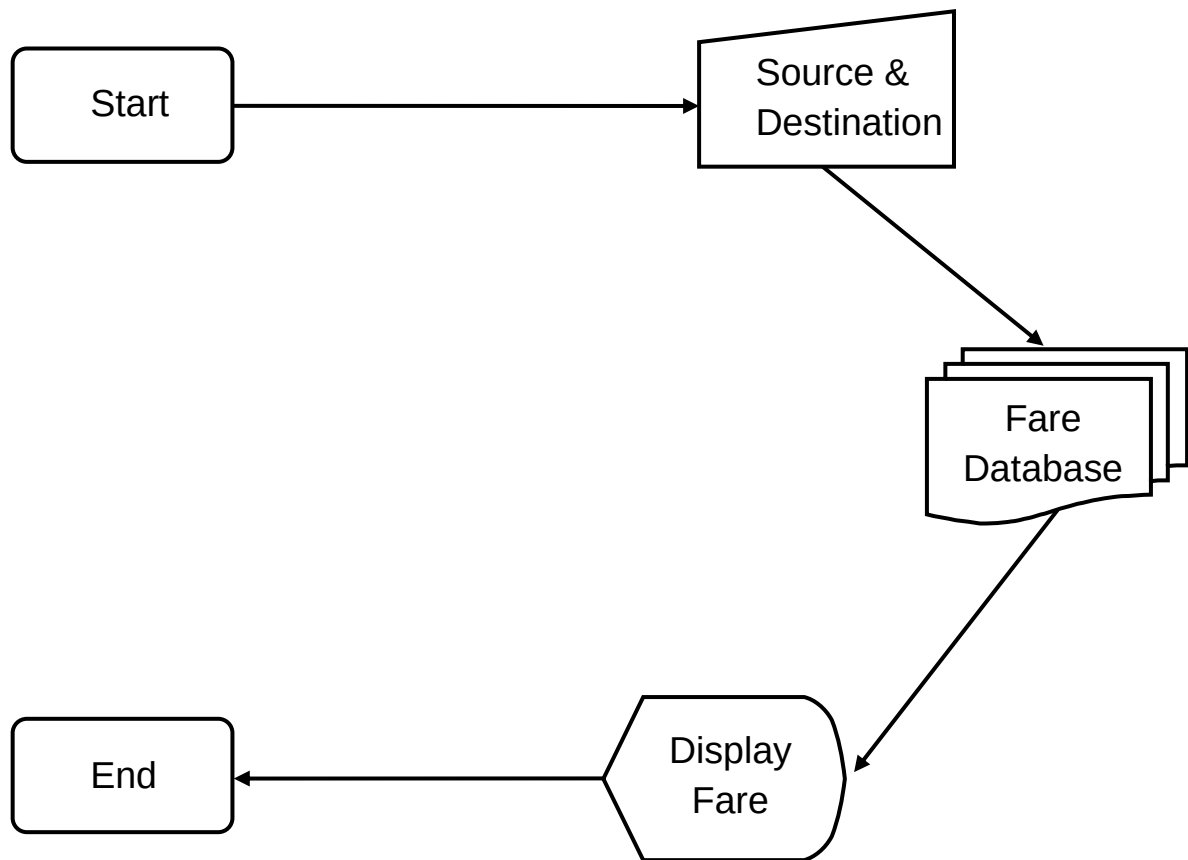


Figure 7: Architecture of Fare Details Portal

Figure 7 indicates the design of Fare and Freight Module. First of all, Source and Destination will be fed to the system, it will retrieve details of fare if data already exist.

3.3. Class Diagram

The Class Diagram which is shown in Figure 8 provides an overview of the, “Excise and Taxation Management System”. The classes, objects and the relation between them is shown in the said figure.

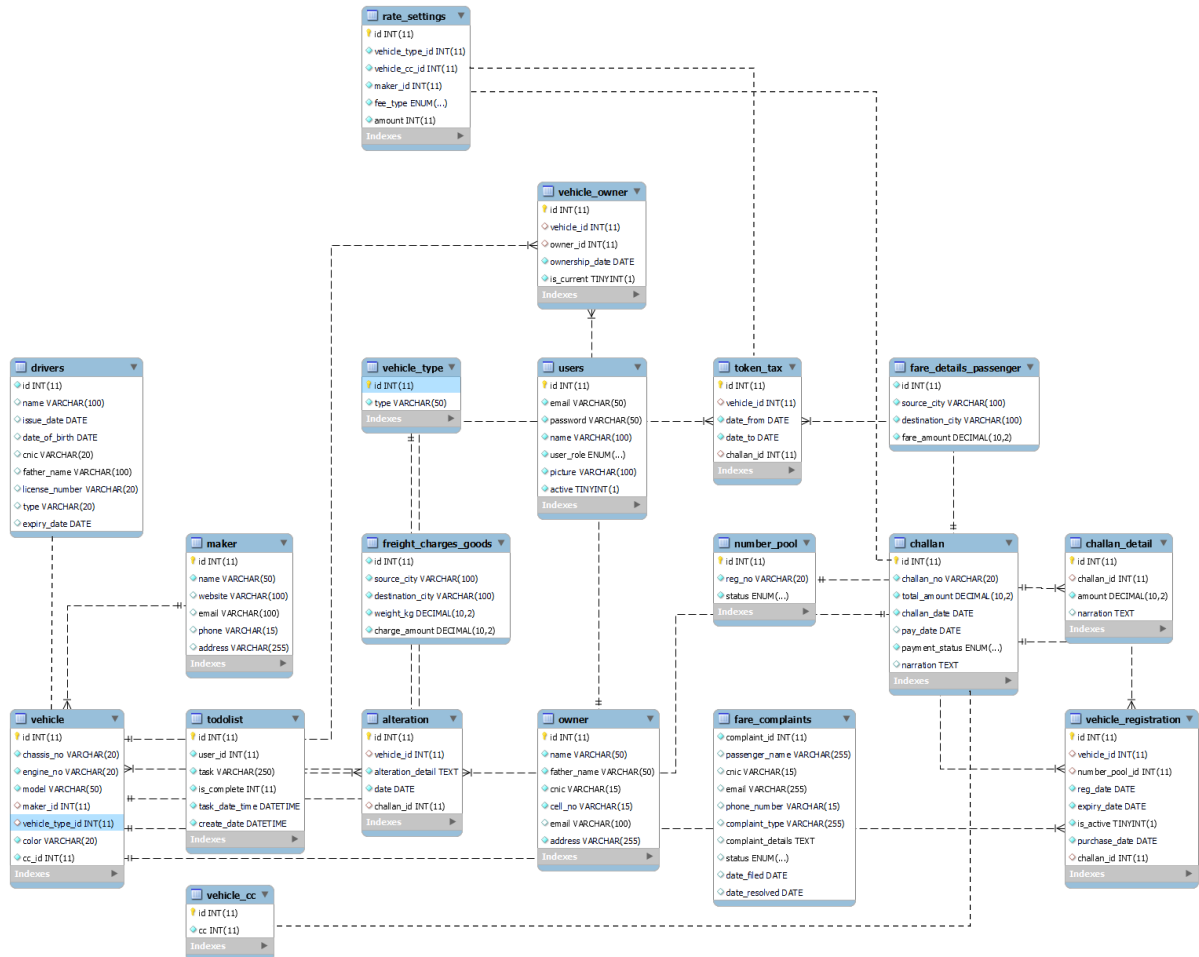


Figure 8: Class Diagram of Excise and Taxation Management System

3.4. ER Diagram

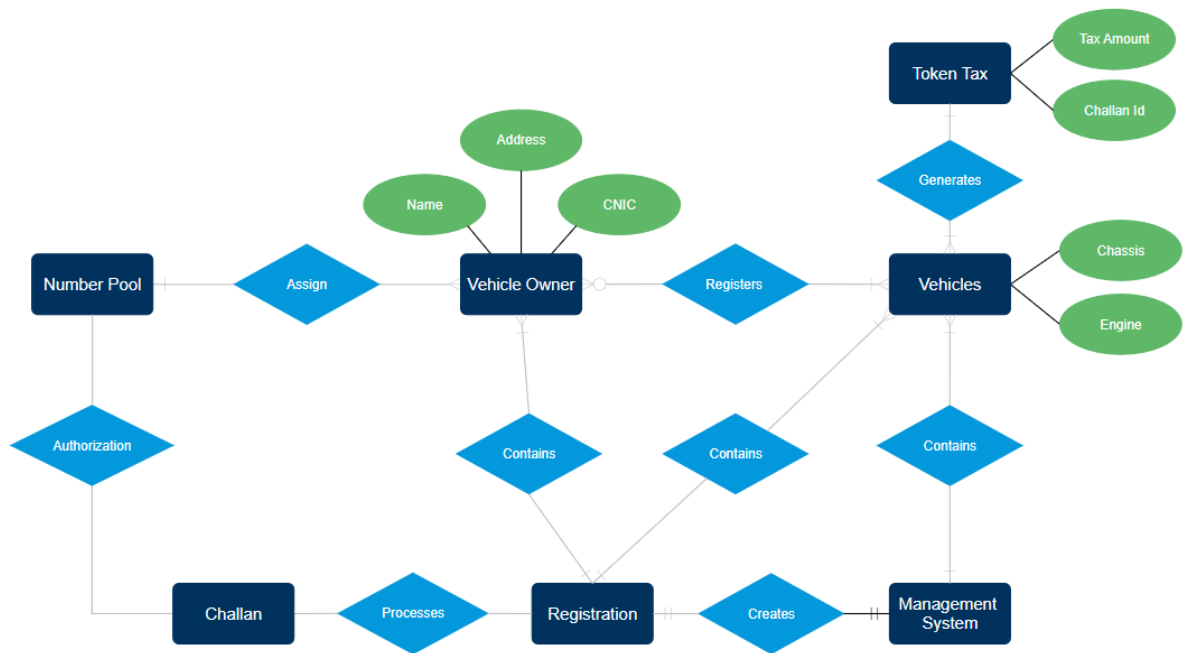


Figure 9: ER Diagram of NCP Vehicle Management

3.5. Data Flow Diagram

3.5.1. Data Flow Diagram of “Excise and Taxation Management”

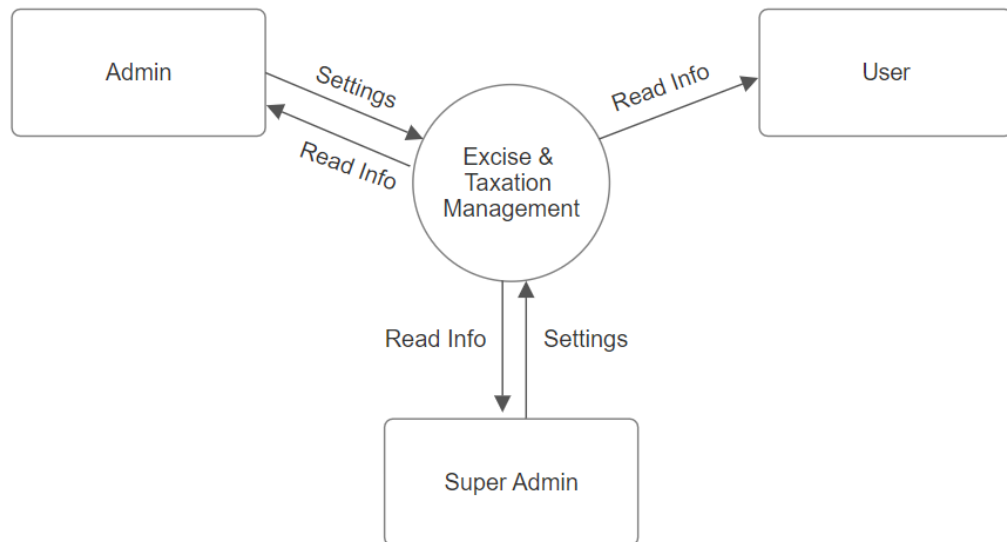


Figure 10: Level-0 DFD of EXCISE AND TAXATION MANAGEMENT

3.5.2. Data Flow Diagram of “Vehicle Registration”

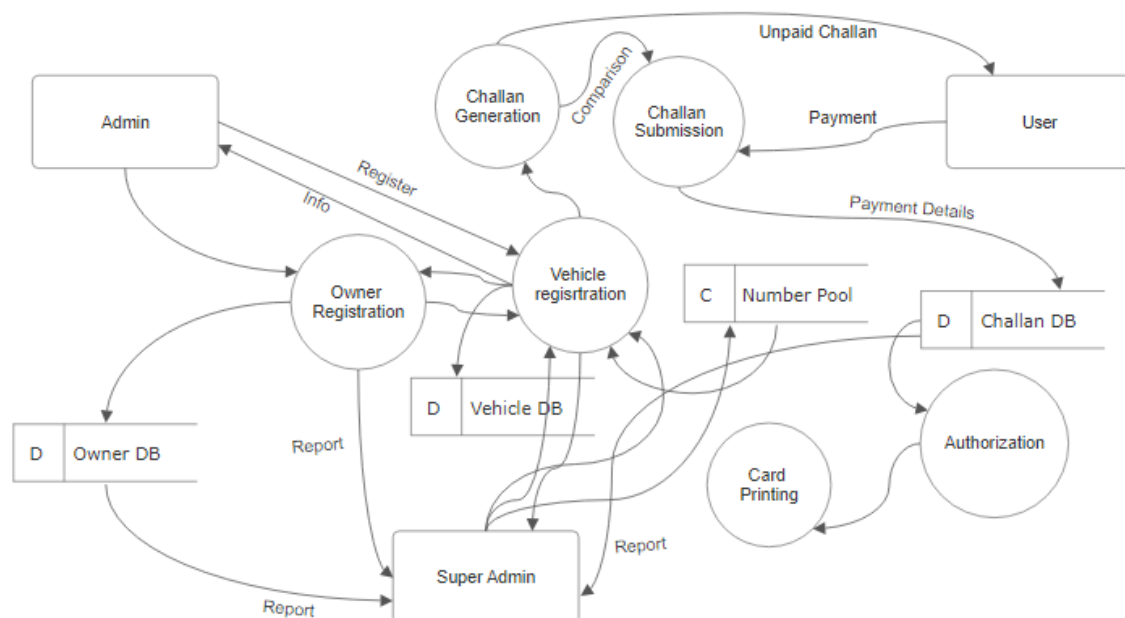


Figure 11: Level-1 DFD of Vehicle Registration

3.5.3. Data Flow Diagram of “Alteration in Vehicles”

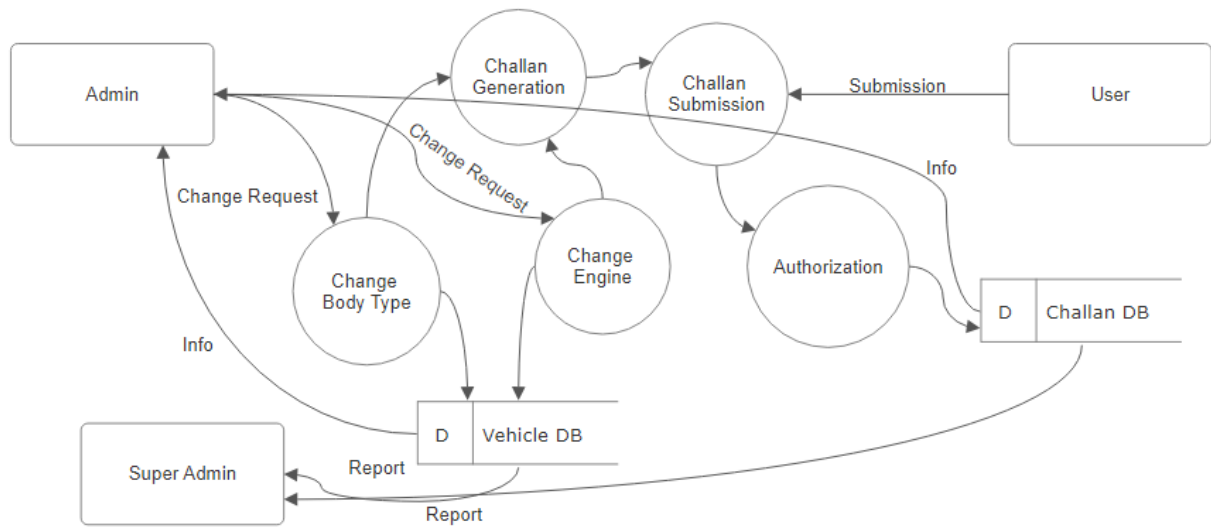


Figure 12: Level-1 DFD of Alteration in Vehicles

3.5.4. Data Flow Diagram of “Verification of Vehicles”

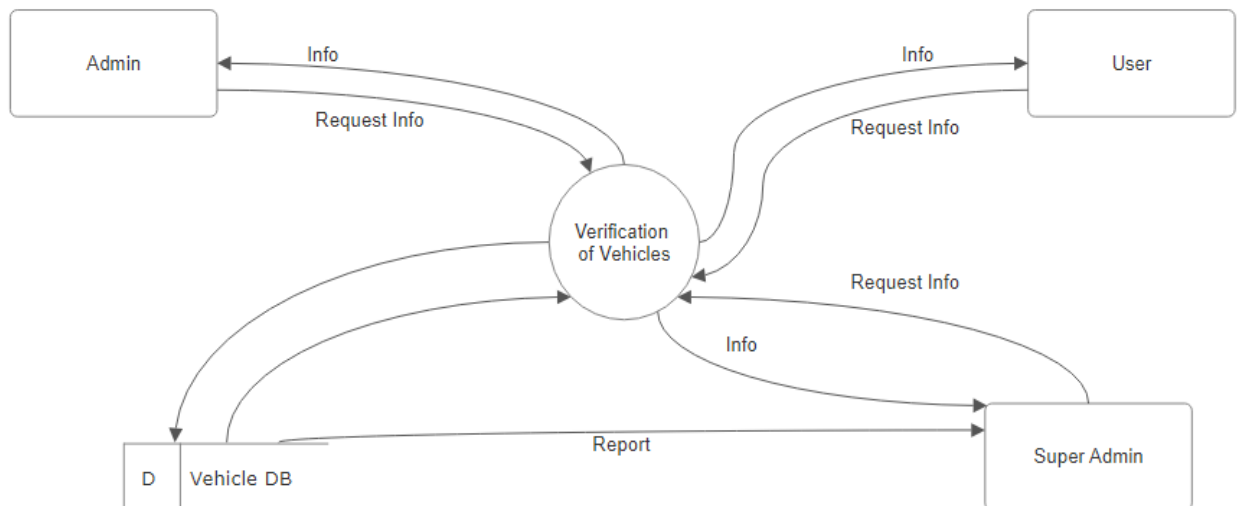


Figure 13: Level-1 DFD of Verification of Vehicles

3.5.5. Data Flow Diagram of “Change of Ownership”

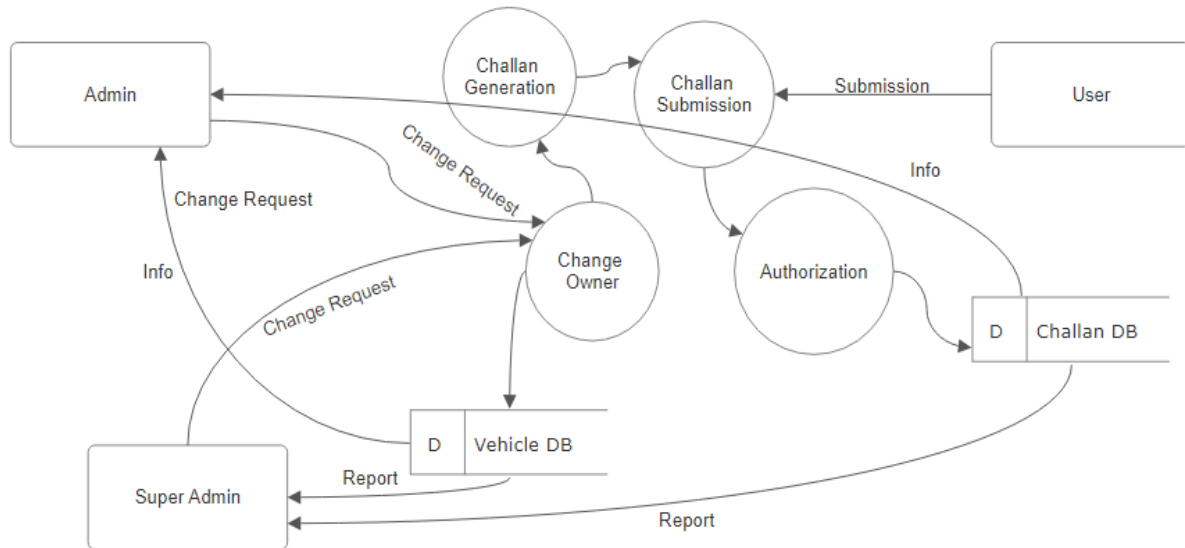


Figure 14: Level-1 DFD of Change of Ownership

3.5.6. Data Flow Diagram of “Company Registration”

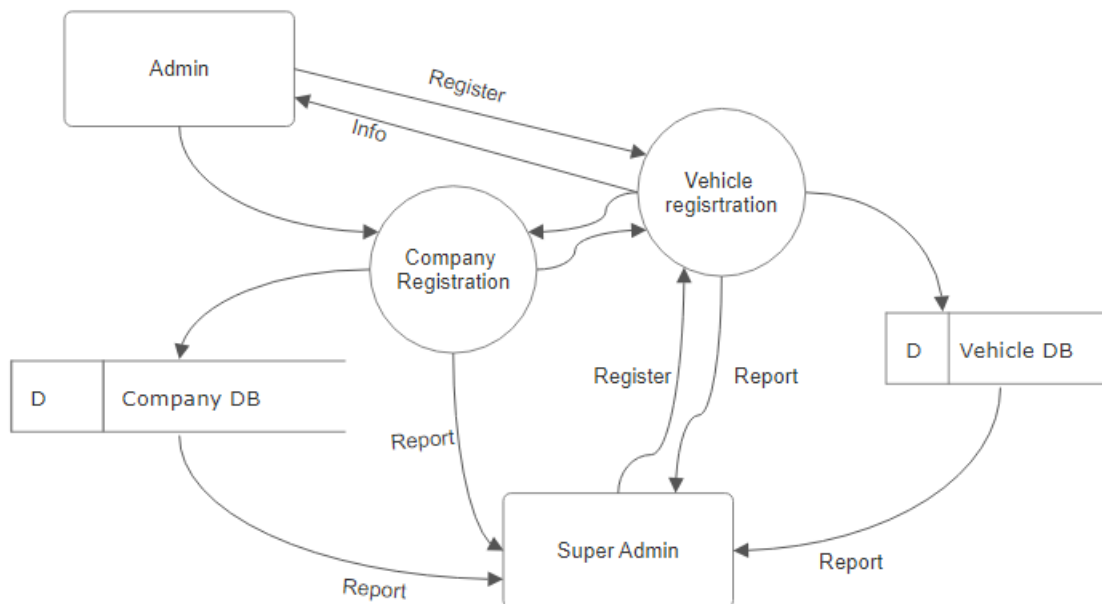


Figure 15: Level-1 DFD of Company Registration

3.5.7. Data Flow Diagram of “Company’s Vehicle Registration”

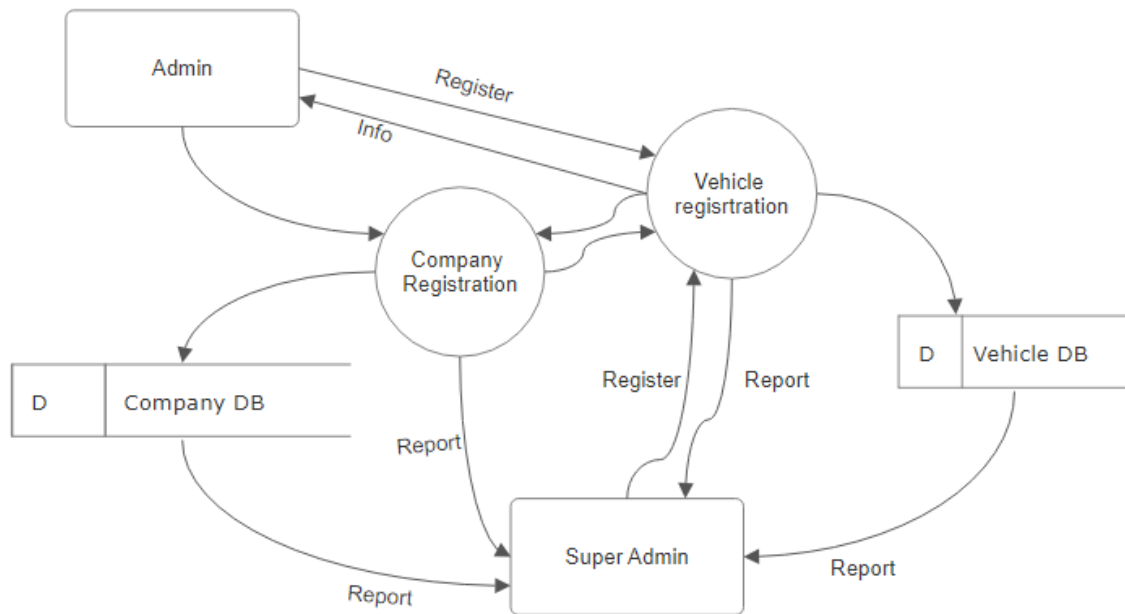


Figure 16: Level-1 DFD of Company’s Vehicle Registration

3.5.8. Data Flow Diagram of “Verification of Driver’s License”

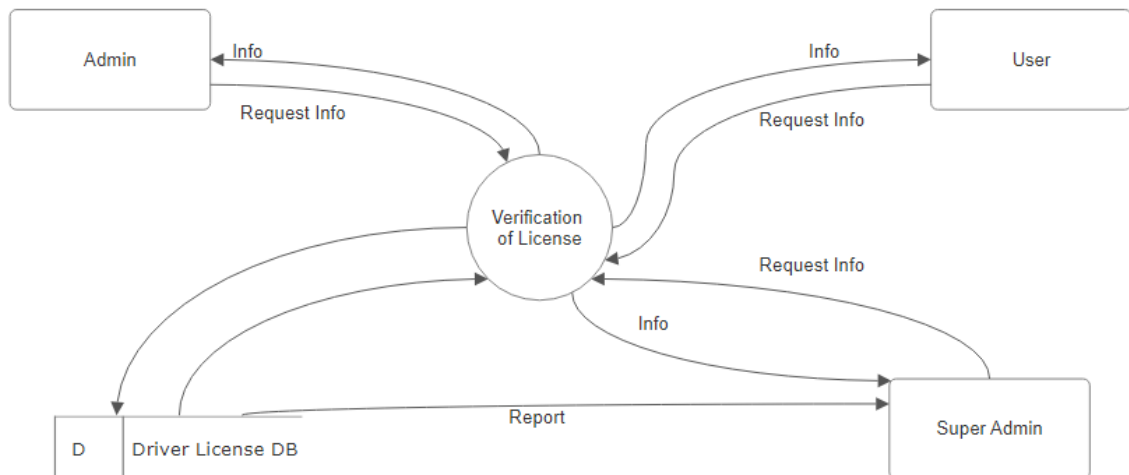


Figure 17: Level-1 DFD of Drivers' License Verification

3.5.9. Data Flow Diagram of “Fare and Freight Details”

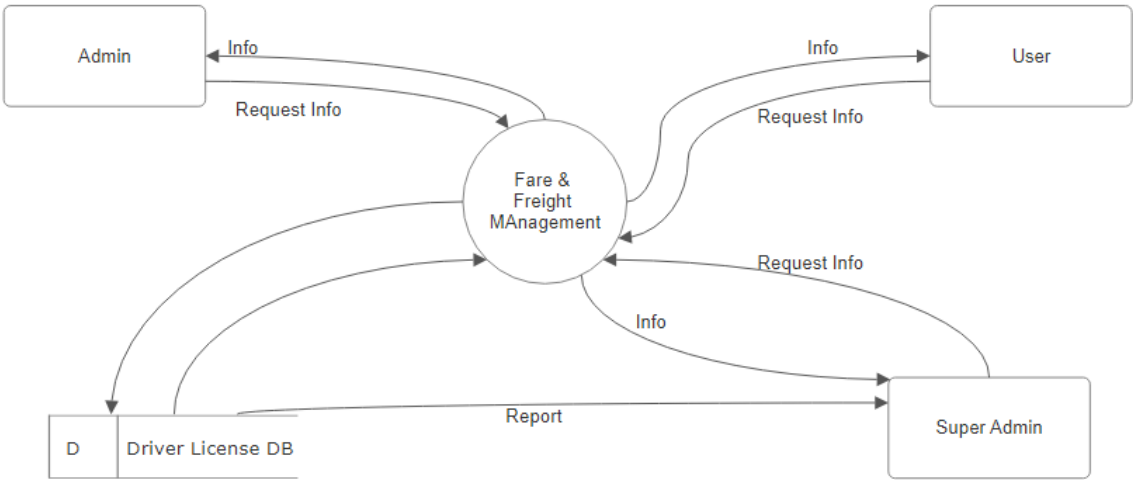


Figure 18: Level-1 DFD of Fare and Freight Details

3.5.10. Data Flow Diagram of “Narcotics License Management”

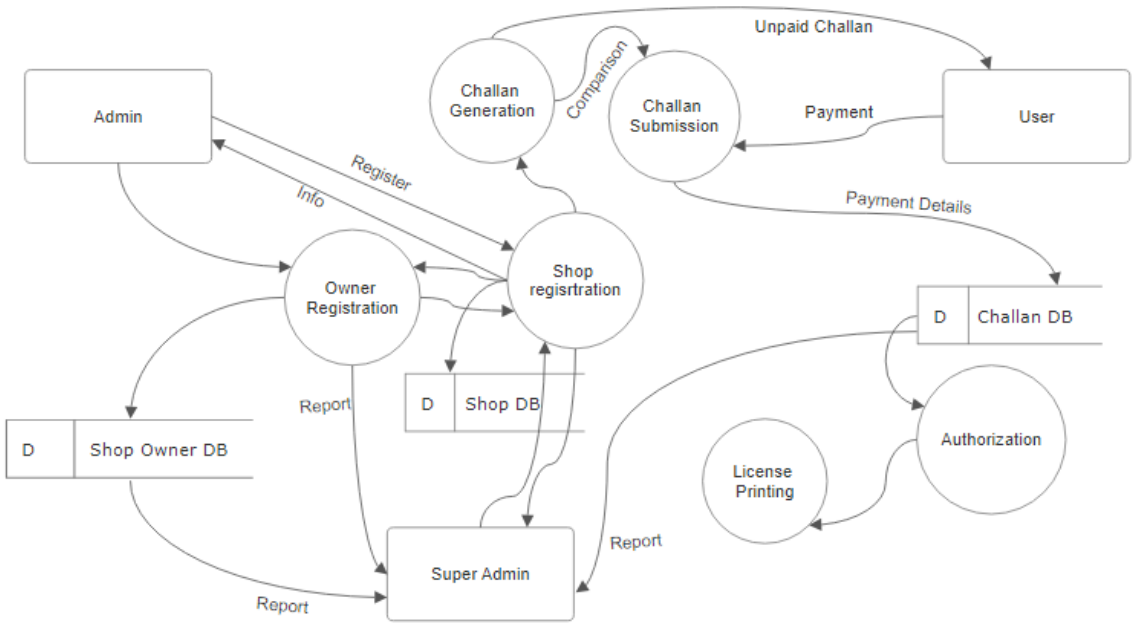


Figure 19: Narcotics License Management

3.6. Use Case Diagram

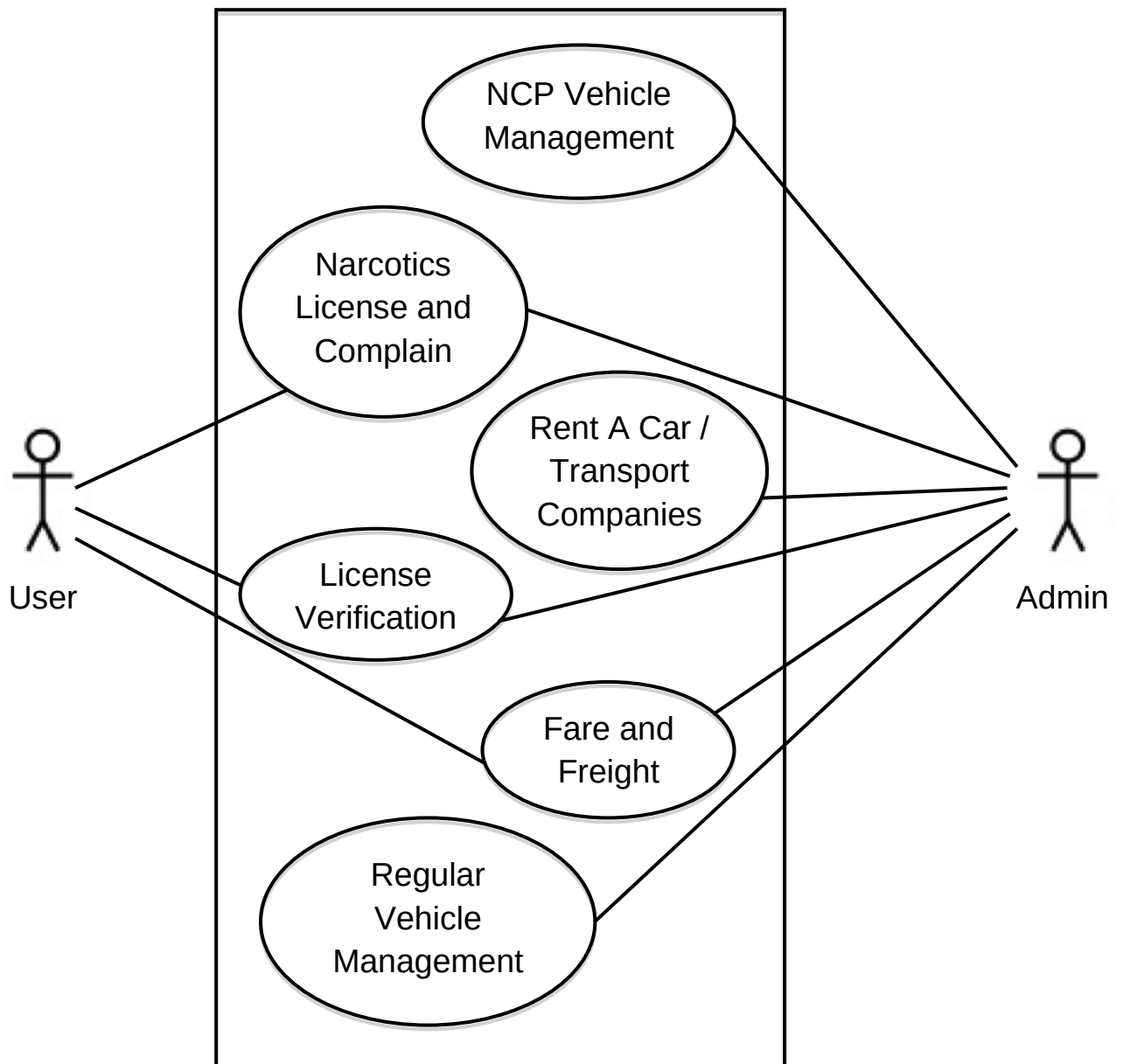


Figure 20: Use Case Diagram of the System

For now, the user has NO access for any of the modules, but Figure 20 shows the depiction after allowing users to access it.

UC-1:

The Admin or Super admin will submit the vehicle registration form. After that, a challan will be generated and handed over to the customer. The customer will bring the challan after paying the fees, which will be authorized after confirming. The vehicle and owner should be registered now.

Table 2: Use Case UC-1

Use Case ID:	UC-1		
Use Case Name:	NCP Vehicle Management		
Created By:		Last Updated By:	
Date Created:		Last Revision Date:	
Actors:	Admin and Super Admin		
Description:			
Trigger:	Clicking on submit button after filling vehicle registration form.		
Preconditions:	The current vehicle must not be registered before.		
Post conditions:	The vehicle and owner should be registered now.		
Normal Flow:	Admin or Super admin will submit the form. After that a challan will be generated and handed over to the customer. The customer will bring the challan after paying the fees, which will be authorized after confirming.	System	
Alternative Flows:			
Exceptions:			

3.7. Functional and Non-Functional Requirements

3.7.1. Functional Requirements:

3.7.1.1. Vehicle Registration:

- Users should be able to register new vehicles in the system.
- Required information includes vehicle details such as chassis number, engine number, make, model, type, year of manufacture, color, and date of registration.
- The system should allow for multiple owners (not at a time) to be associated with a single vehicle.
- Calculation of tax on a quarterly basis and generation of token tax for the current fiscal year should be automated.
- Generation of treasury slips (TR6) with unique challan numbers for tax payments.
- Option to receive payment confirmation against the given challan number.
- Authorization process for vehicle registration.
- Printing and delivery of registration cards.

3.7.1.2. Vehicle Alteration:

- Ability to change attributes of a vehicle, such as engine or body type, with applicable fees.
- Generation of treasury slips (TR6) for alteration fees.

3.7.1.3. Online Verification of Vehicles:

- Posting of vehicle data on a live interface for public verification.
- Downloading of vehicle data in specified formats from the database.

3.7.1.4. Transfer of Ownership:

- Capability to transfer vehicle ownership with new owner details including CNIC, name, and other relevant information.
- Generation of treasury slips (TR6) based on vehicle type and specifications.
- Annual Registration Renewal/Token:
- Renewal of vehicle registration and token tax on an annual basis.

- Notification system for registration expiry.

3.7.1.5. Database Management System for Rent A Car/Transport Companies:

- Registration of rent-a-car and transport companies.
- Application process for No Objection Certificates (NOC) for rent-a-car companies.
- Submission and verification of company NOCs on the portal.
- Online vehicle registration for companies.
- Submission and renewal of route permits, registration, fitness, and insurance of vehicles on the portal.
- Verification of vehicles, route permits, fitness, and insurance online.
- Expiry notifications for route permits, fitness, and insurance of vehicles.
- Asynchronous mobile application for check-posts to ensure updated data.

3.7.1.6. Database Management System for Normal (Regular) Vehicles:

- Vehicle registration for regular vehicles.
- Online verification of regular vehicles.
- Transfer of ownership for regular vehicles.
- Annual renewal/token tax for regular vehicles.
- Notification system for renewal expiry of regular vehicles.

3.7.1.7. Verification of Driver License:

- Notification system for driver's license expiry on the Excise's Dashboard.

3.7.1.8. Narcotics License and Complain Portal:

- Application process for tobacco vendor licenses.
- Renewal process for tobacco vendor licenses.
- Listing of warned and suspended vendors.
- Vendor verification system.
- Narcotics complaint management system.

3.7.1.9. Fare Details Portal:

- Display of fare details.
- Calculation of freight charges.

- Fare complaint management system.

3.7.2. Nonfunctional Requirements:

3.7.2.1. UI / UX:

- The system should have a user-friendly interface that is easy to navigate. The user interface is far better than the system the possess currently. But, it can be improved more after judging user experience.

3.7.2.2. Performance:

- The proposed system is capable of handling a large volume of registration and verification requests without significant slowdowns.

3.8. User Interface Design

We tried our best to provide our user a simple and easy-to-use user interface. It is way better than the previous one they already have. But there is always possibility of betterment, so our UI can also be improved after judging user experience.

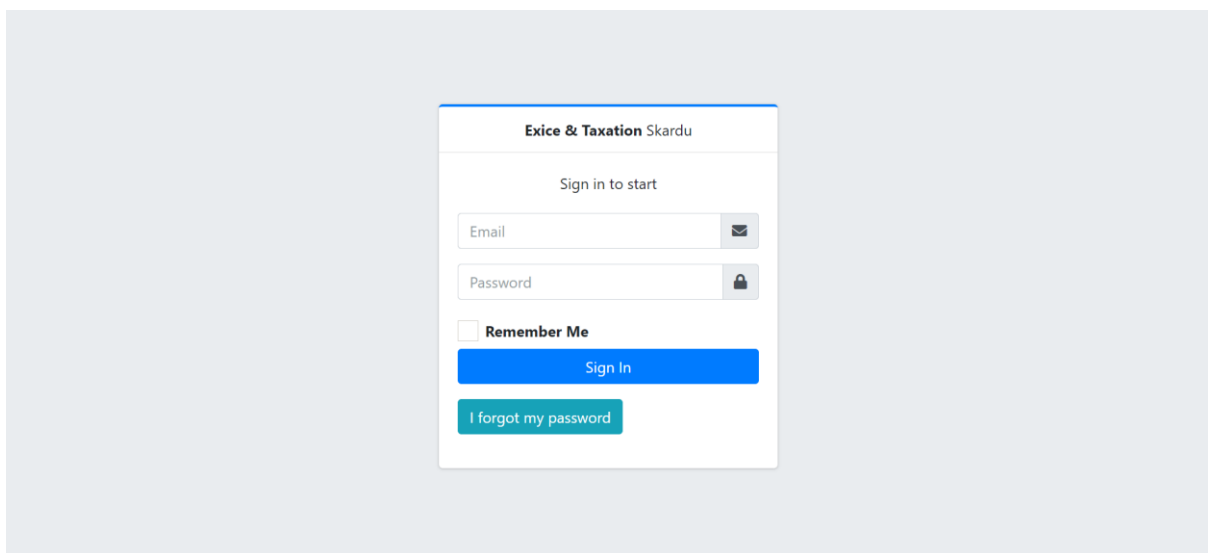


Figure 21: Login View

Figure 21 shows the login page, from where the user has to put his credentials.

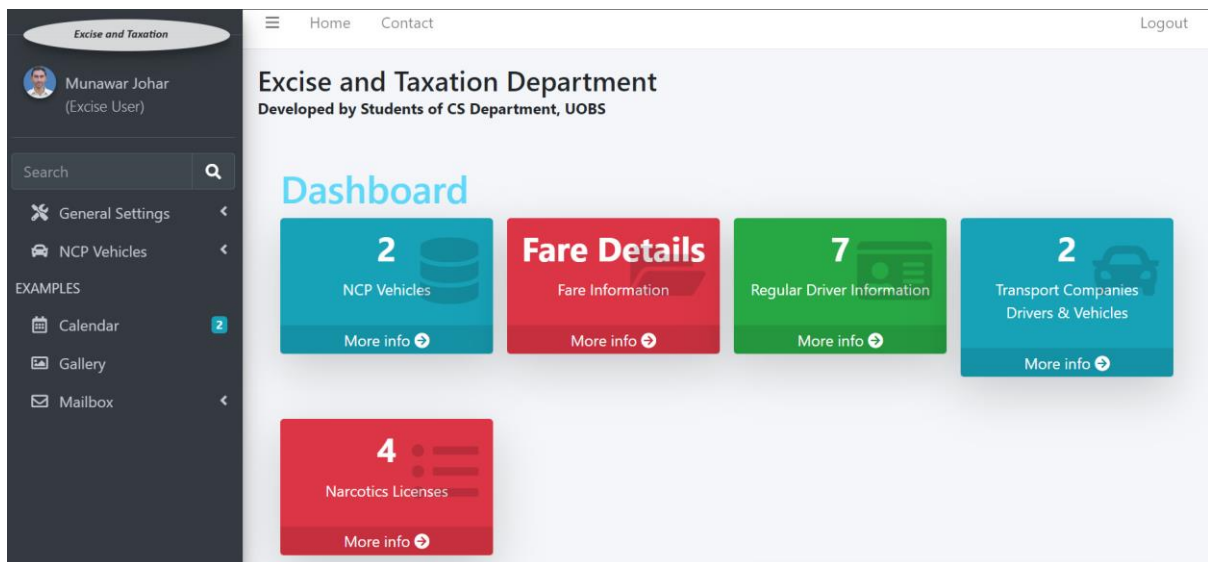


Figure 22: Homepage UI

Figure 22 shows the Homepage of Excise and Taxation Management System. We can see 3 modules on the homepage right now. Rest of the modules will also be added here, so it can be navigated easily. We have completed some of them but did not manage to put it on homepage yet.

Expired Driver Licenses									
Copy CSV Excel PDF Print Column visibility					Search:				
S.No	Name	Issue Date	Date of Birth	CNIC	Father's Name	License Number	Type	Expiry Date	Edit
1	Munawar Hussain	2020-02-02	1999-07-14	72303-0343264-7	Rajab	GBEXCISE-148	LT	2024-03-05	Action
2	Munawar Johar	3333-03-31	2000-12-12	71101-2345678-1	Rajab	BLN400	333	2022-03-03	Action
S.No	Name	Issue Date	Date of Birth	CNIC	Father's Name	License Number	Type	Expiry Date	Edit

Showing 1 to 2 of 2 entries

Previous 1 Next

Figure 23: UI, showing Expired Licenses of Drivers

Figure 23 shows the expired licenses of drivers. It can be seen in the figure that the design is simple yet easy to navigate.

Some other examples of UI are attached, so that a clear picture of UI sets into readers' mind.

Excise and Taxation

Mubarak Ali
(Excise User)

Search

Q

NCP Vehicles

Fare portal

Drivers Licenses

Drivers Details

Expired Licenses

Verify Licenses

Expiry Soon Licenses

Images

EXAMPLES

Calendar

Gallery

Add Add/Update Driver

Driver Licenses

Copy CSV Excel PDF Print Column visibility

Search:

S.No	Name	Issue Date	Date of Birth	CNIC	Father's Name	License Number	Type	Expiry Date	Edit
1	Munawar Johar	3333-03-31	2000-12-12	71101-2345678-1	Rajab	BLN400	333	2022-03-03	Action
2	Kamal Ali	2020-12-12	1999-12-09	71101-2345678-1	Ali Hasan	GBEXCISE-150	HTV	2024-05-02	Action
3	Munawar Hussain	2020-02-02	1999-07-14	72303-0343264-7	Rajab	GBEXCISE-148	LT	2024-03-05	Action
4	kamal	2222-02-22	2022-03-31	6373	Abid	BLN-500	333	2025-04-04	Action
5	Munawar Hussain	2024-04-17	1999-02-02	71101-2345678-1	Rajab	GBEXCISE-148	HTV	2025-04-17	Action

Figure 24: UI, showing list of drivers' licenses

Excise and Taxation Department

Developed by Students of CS Department, UOBS

Add Add/Update Complaint

Complaints

Copy CSV Excel PDF Print Column visibility

Search:

S.No	Passenger Name	CNIC	Email	Phone Number	Complaint Type	Complaint Details	Status
1	MunawarJohar	71101-2345678-1	munawarjohar148@gmail.com	03456789112	Fare	This is new	Closed
2	Munawar Kamal	71101-2345678-1	munawar151512@gmail.com	03456789112	Fare	This is	Closed
3	MunawarJ ohar	71101-2345678-1	munawarjohar148@gmail.com	03456789112	Fare new	this new message	Open
4	MunawarJ ohar	71101-2345678-1	munawar151512@gmail.com	03456789112	Fare	this is new	Open

Figure 25: UI, showing list of complaints

Excise and Taxation

Mubarak Ali
(Excise User)

Search

NCP Vehicles
Fare portal
Passenger fare Details
Freight Charges Details
Passenger Charges
Freight Charges
Complaints Details
Drivers Licenses
EXAMPLES
Calendar
Gallery

Developed by Students of CS Department, UOBS

Add Add/Update Charge

Charge Details

Copy CSV Excel PDF Print Column visibility
Search:

S.No	Source City	Destination City	Charge Amount	Weight (kg)	Edit
1	Skardu	Roundu	5.00	1.00	Action
2	Skardu	Kharmang	10.00	1.00	Action
3	Gitgit	Hunza	30.00	1.00	Action
4	Islamabad	Skardu	50.00	1.00	Action
S.No	Source City	Destination City	Charge Amount	Weight (kg)	Edit

Showing 1 to 4 of 4 entries

Previous 1 Next

Figure 26: UI of Fare and Freight Portal

Excise and Taxation

Mubarak Ali
(Excise User)

Search

General Settings
NCP Vehicles
Drivers Licenses
Fare Portal
EXAMPLES
Calendar
Gallery
Mailbox

Add Add/Update Complaint

Passenger Name:

CNIC:

Email:

Phone Number:

Complaint Type:

Complaint Details:

Figure 27: UI of Complaint Form for Fare Module

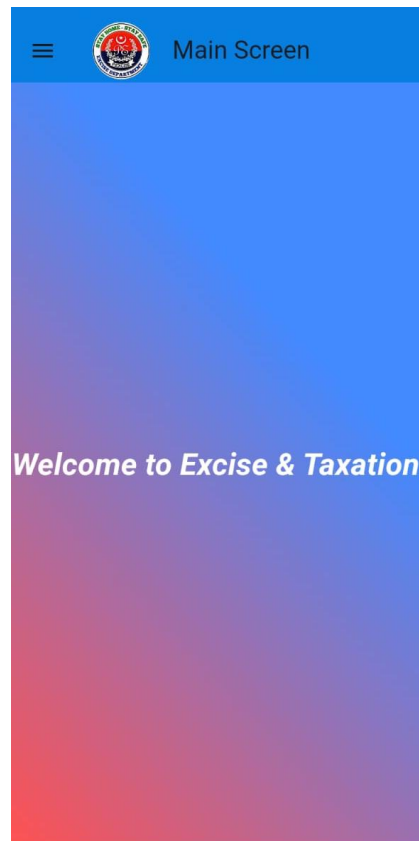


Figure 28: Mobile Application (Home Page)

Figure 29: Fair Complaint Portal

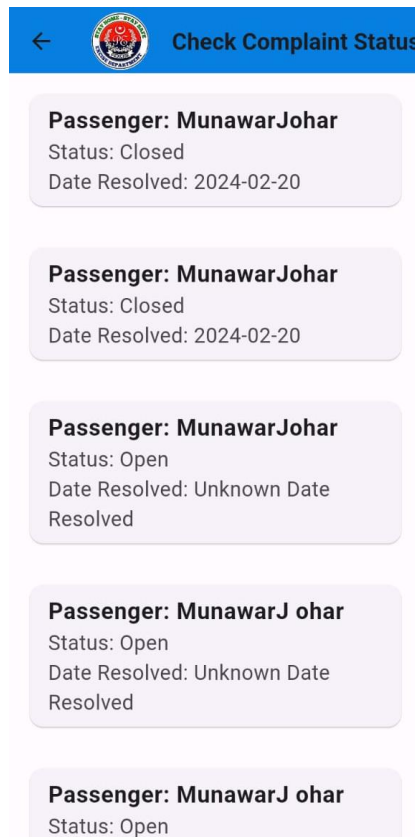


Figure 30: Complaint Status

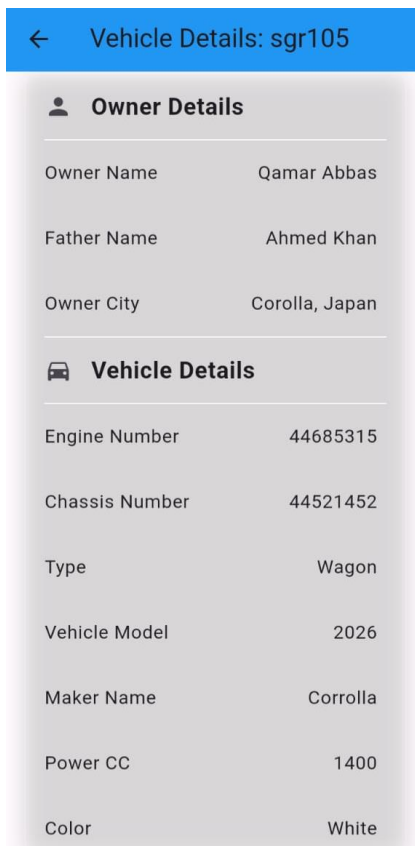


Figure 31: Vehicle's Details

Chapter 4

System Implementation

Building on the ideas and plans discussed in previous chapters, this chapter examines Excise and Taxation project. Here, we will examine the intricate inner workings of the platform's development, from inception to completion and integration of all of its numerous parts. This chapter will give a thorough rundown of the implementation procedure, emphasizing the technical considerations and difficulties that were faced by us. In addition to showing how concepts can be effectively developed into a working platform, we will also cover any unanticipated challenges that may come up while the project is in the early stages. We want to present an understandable and thorough overview of the project's execution by pointing out these issues and highlighting the solutions put in place. The time commitment required for addressing typical implementation difficulties such problems will be covered in this chapter.

4.1. System Simulation / System Development Process

When we were building the Excise and Taxation Management System for Baltistan, we followed a step-by-step process to make sure everything worked smoothly. We did this using the following steps.

4.1.1. Development Environment:

To write and test our code, we used a special software called Visual Studio Code, which helped us write code faster and catch mistakes. The best part of VS Code is that it suggests possible code before we write it. It helped us to code faster.

4.1.2. Packages and Libraries:

To make things easier, we used some ready-made tools and pieces of code that other people had already created. For example, we used some PHP utilities to build the main parts of our website, and another tool called Bootstrap to make it look nice and work well on different devices. We also used a tool like jQuery to help with certain tasks on the website.

By using these tools and following these steps, we were able to build the Excise and Taxation Management System for Baltistan in a way that was efficient, organized, and easy to work with.

4.2. Software Description

This project is all about simplifying and modernizing how the Excise and Taxation Department of Baltistan handles important tasks.

The Problem: Right now, the department uses a lot of manual paperwork and old-fashioned methods to manage things like car rental services, transport companies and tobacco vendor's details. Surprisingly, they have all data of NCP vehicles in papers and files. All they have is 19% of that data is shifted in MS Excel. This causes a bunch of issues, like less revenue generation, missed deadlines, lost money, and trust problems.

Our Approach: We're creating a management system that will put all the necessary information in one place, making it way easier to keep things organized. This system will help remind everyone when important documents, like permits, are about to expire so they can renew them in time. It will also give the department the data they need to make smart decisions.

Why It Matters: This project is important because it makes things run more smoothly in Baltistan. It saves time, helps the department collect the money they're owed, and makes sure that everyone can see how things are being done fairly. Plus, it's in line with the rest of the world, which is going digital, and it can protect our environment by controlling the number of vehicles on our roads.

In simple terms, this project is all about making life easier and better for everyone in Baltistan.

4.2.1. Major Modules

It is to be noted that public cannot access this website for now. The department is not willing to give its access to public. They will allow public to access it later.

4.2.1.1. Database management for NCP vehicles.

This module deals with keeping track of vehicles that haven't paid customs. They are currently storing the data of NCP vehicles into hard form. They need a management system to handle this issue. This module will help them maintaining a robust database. They will also be able to differentiate which car's token tax is paid and which car's token tax is not paid ensuring collection of higher revenue for them.

```
foreach($registrations as $registration)
{
    $status = "Pending";
    if($registration['is_active']==1)
    {
        $status = "Registered";
    }
    $sno++;
    $registration_id = $registration['id'];
    $vehicle_id = $registration['vehicle_id'];
    $number_pool_id = $registration['number_pool_id'];
    $challan_id = $registration['challan_id'];
    $where_vehicle = "id = '$vehicle_id'";
    $vehicle = getSingleRow("vehicle","id, chassis_no, engine_no",$where_vehicle);
    $where_pool = "id = '$number_pool_id'";
    $vehicle_pool = getSingleRow("number_pool","id,reg_no",$where_pool);
    $where_challan = "id = '$challan_id'";
    $vehicle_challan = getSingleRow("challan","id,challan_no,total_amount",$where_challan);
}
```

Figure 32: Code snippet of NCP Management Module

Description: This piece of code will display the details of registered vehicles.

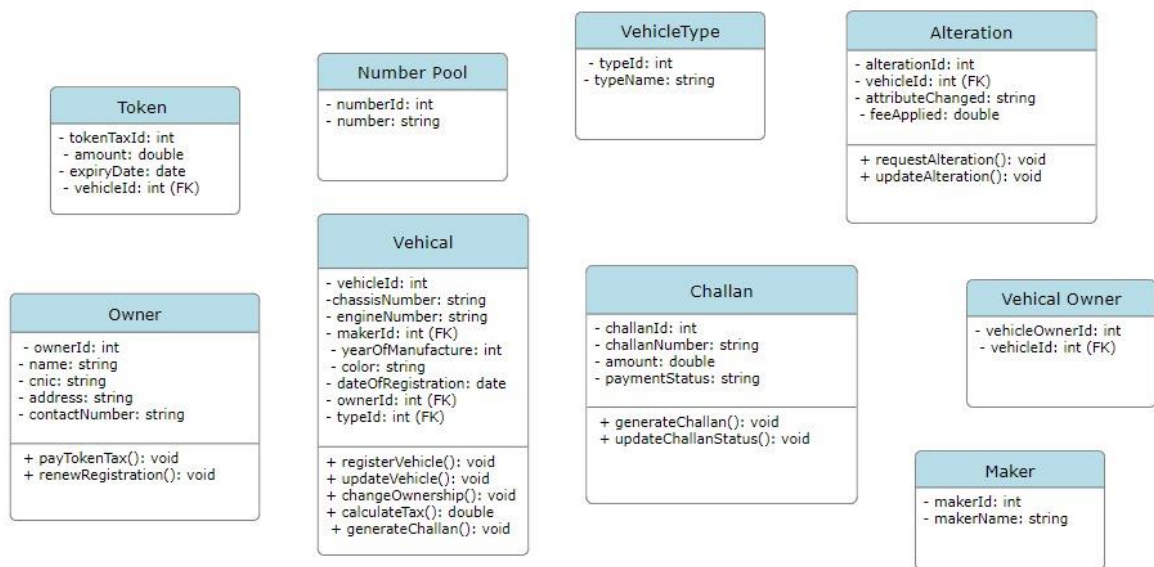
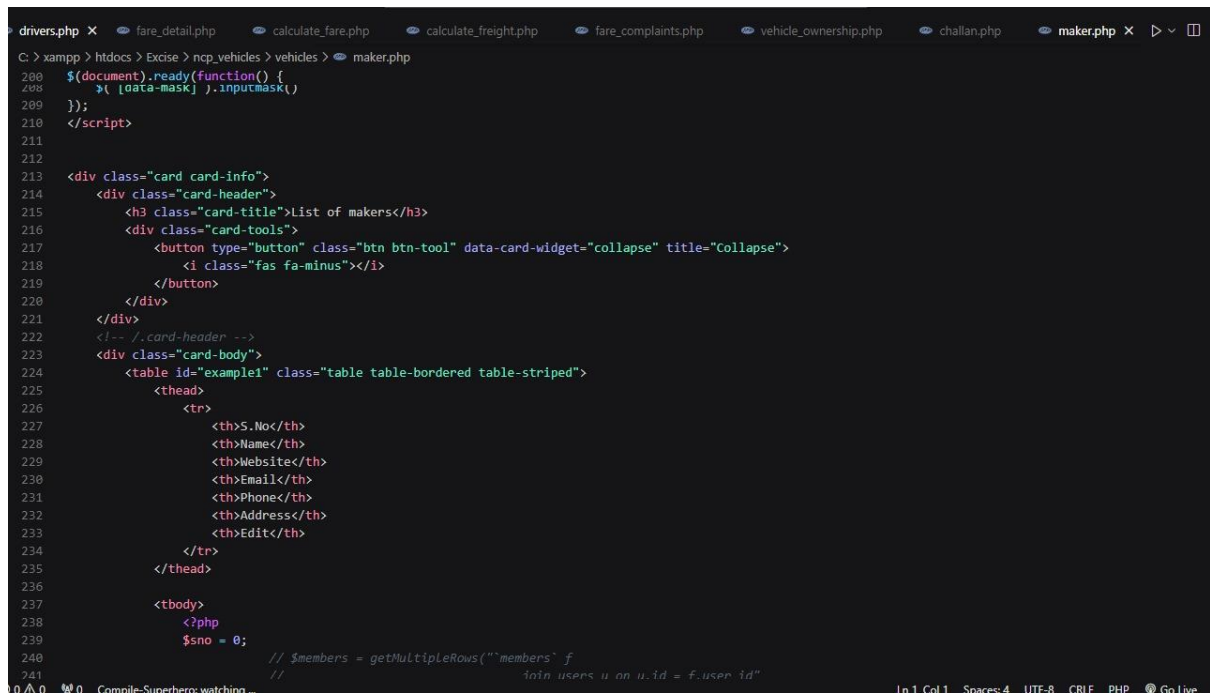


Figure 33: Class Diagram of NCP Vehicle Management Module

4.2.1.2. Database management for normal (Regular) vehicles.

This module handles the registration and upkeep of regular vehicles. It helps with registration, checking ownership, renewing licenses, and reminding owners when they need to renew.



```
drivers.php x fare_detail.php calculate_fare.php calculate_freight.php fare_complaints.php vehicle_ownership.php challan.php maker.php x
C:\> xampp > htdocs > Excise > ncp_vehicles > vehicles > maker.php
200 $(document).ready(function() {
201     $('#data-mask').inputmask();
202 });
203 </script>
204
205 <div class="card card-info">
206     <div class="card-header">
207         <h3 class="card-title">List of makers</h3>
208         <div class="card-tools">
209             <button type="button" class="btn btn-tool" data-card-widget="collapse" title="Collapse">
210                 <i class="fas fa-minus"></i>
211             </button>
212         </div>
213     </div>
214     <!-- /.card-header -->
215     <div class="card-body">
216         <table id="example1" class="table table-bordered table-striped">
217             <thead>
218                 <tr>
219                     <th>S.No</th>
220                     <th>Name</th>
221                     <th>Website</th>
222                     <th>Email</th>
223                     <th>Phone</th>
224                     <th>Address</th>
225                     <th>Edit</th>
226                 </tr>
227             </thead>
228             <tbody>
229                 <?php
230                 $sno = 0;
231                 // $members = getMultipleRows("members" f
232                 //         join users u on u.id = f.user id"
233             </tbody>
234         </table>
235     </div>
236 </div>
```

Figure 34: Code Snippet of Normal Vehicles

Figure 30 illustrates the display of drivers' information on screen. It can be seen that it is obtained using table in HTML. It is to be noted that the requirement has changed and we are currently not working on this module.

4.2.1.3. Verification of Driver's License.

This module helps to check if driver licenses are still valid and sends reminders when they're about to expire.

```
<?php if (!empty($drivers_expiry_soon)) : ?>
    <div class="table-responsive">
        <table class="table table-bordered">
            <thead>
                <tr>
                    <th>Driver Name</th>
                    <th>Father's Name</th>
                    <th>CNIC</th>
                    <th>Date of Birth</th>
                    <th>License Number</th>
                    <th>Type</th>
                    <th>Issue Date</th>
                    <th>Expiry Date</th>
                </tr>
            </thead>
            <tbody>
                <?php foreach ($drivers_expiry_soon as $driver) : ?>
```

Figure 35: Code Snippet of licenses expiring within a month

Codes written in Figure 31 will display the drivers' list, whose licenses are going to expire within a month.

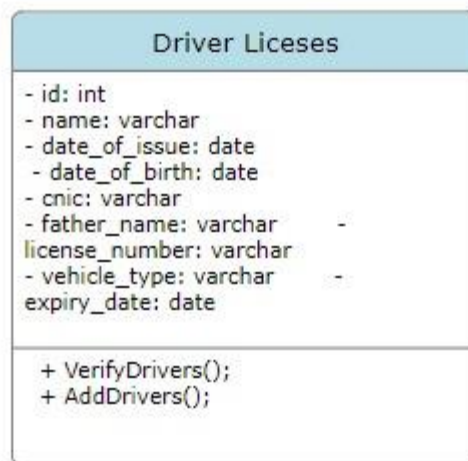


Figure 36: Class Diagram of License Verification Module

4.2.1.4. Fare details portal.

This deals with sharing details about fares. It ensures everyone knows how much they need to pay and helps increasing in transparency regarding fares.

```
// Query to fetch distinct source cities from the database
$sourceCityQuery = "SELECT DISTINCT source_city FROM fare_details_passenger";
$sourceCityResult = mysqli_query($conn, $sourceCityQuery);
while ($row = mysqli_fetch_assoc($sourceCityResult)) {
    echo "<option value='{$row['source_city']}'>{$row['source_city']}</option>";
}
?>
</select>
</div>
<div class="form-group">
<label for="destinationCity">Destination City</label>
<select class="form-control" id="destinationCity" name="destinationCity" required>
    <?php
        // Query to fetch distinct destination cities from the database
        $destinationCityQuery = "SELECT DISTINCT destination_city FROM fare_details_passenger";
        $destinationCityResult = mysqli_query($conn, $destinationCityQuery);
        while ($row = mysqli_fetch_assoc($destinationCityResult)) {
            echo "<option value='{$row['destination_city']}'>{$row['destination_city']}</option>";
        }
        ?>
    </select>
</div>
<button type="submit" class="btn btn-primary">Generate Fare</button>
</form>
```

Figure 37: Fare Generator Code

The fare will be generated once the user enters the source and destination.

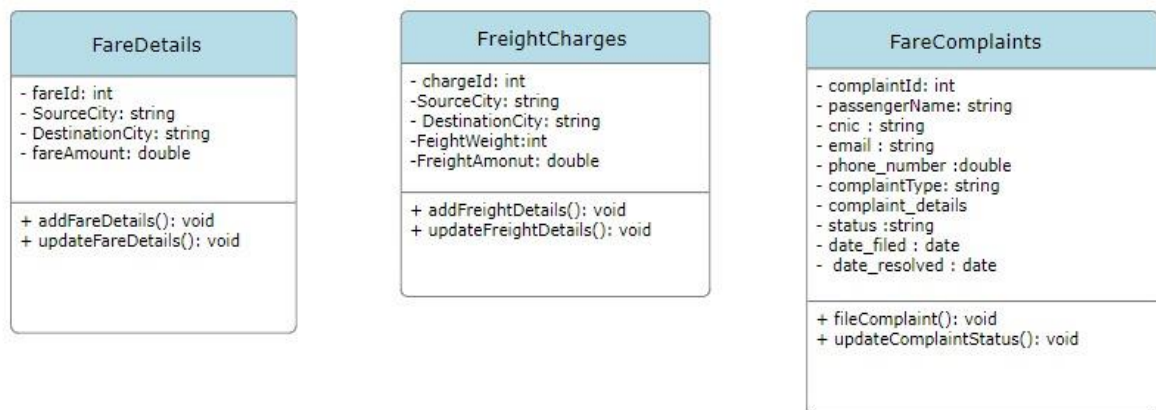


Figure 38: Class Diagram of Fair Details portal

Chapter 5

Testing and Deployment

5.1. Testing Methodology

As we have pre-defined set of features. For this, Unit Testing is suitable. As Unit Testing is used to test each and every feature / module separately, we used Unit Testing methodology.

5.2. Test Cases

During the testing phase, we test each module. We fed some input and judged the result after comparing the expected result and the actual result. According to our testing data, it can be said that our project is up to the mark and meet all the requirements. Here we will discuss all the test cases which we handled.

5.2.1. Test Case Login

- **Test case description:** To test the user credentials, we must enter the right and wrong credentials at least once.
- **Expected result of the test case:** If the credentials are right, access should be granted, otherwise not.
- **Actual result of the test case:** We entered the wrong credentials first so it declined to grant access to the system. Then, we entered the right credentials and it granted access successfully.

Table 3: Test Case Login

Test Case ID: TC-001 Test Type: Unit Testing Component: Login	Test Item: Login Test Case Name: Test Case Login Release Version: 1.0
Test Case Description: To test the user credentials, we must enter the right and wrong credentials at least once.	
Pre-Condition	Post Condition
Login Credentials must be there in the database.	
Input Data/Event	Expected output Data
Email: email@gmail.com Password: 12345	Access should be granted
Test Case Result	
Access granted	

Actual Input Data/Event	Actual Output Data/Event
Email: email@gmail.com Password: 12345	Access granted
Test Case Result	
Test is performed successfully	

Here are the visuals of the test.

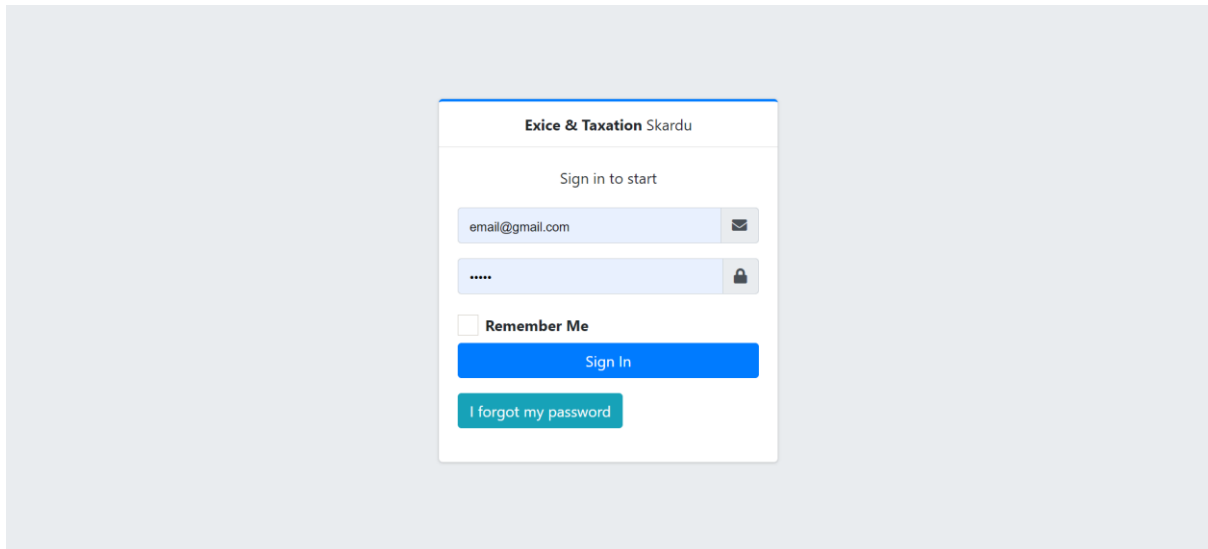


Figure 39: Login View

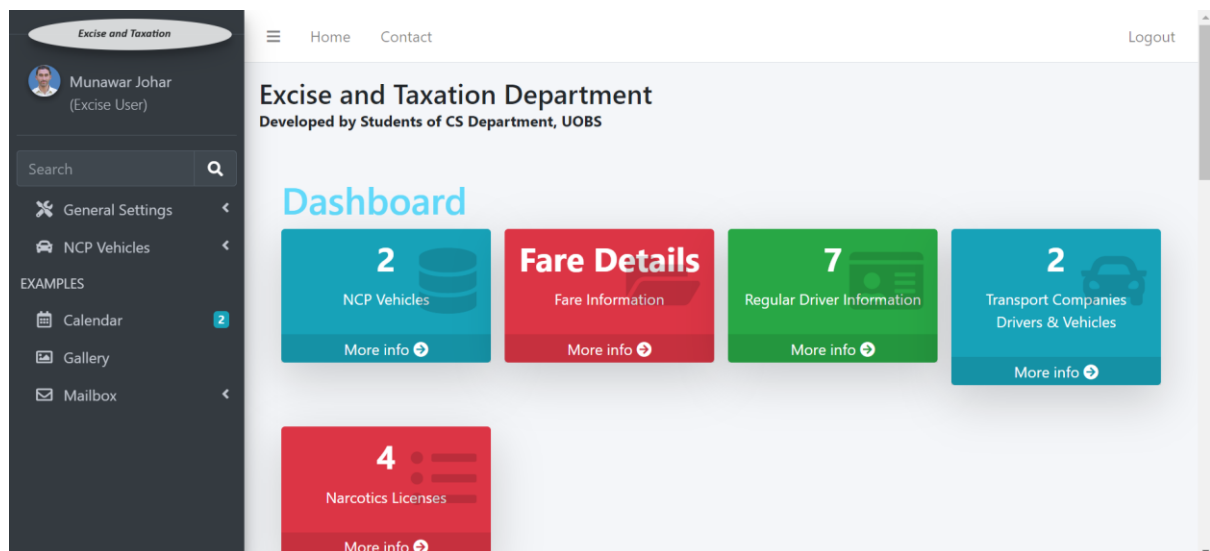


Figure 40: Login Successful (Homepage)

5.2.2. Test Case NCP Vehicle Registration

- **Test case description:** This module includes the registration process of NCP vehicles.

- **Expected result of the test case:** After the registration process, the data should be shown in the vehicles list.
- **Actual result of the test case:** We registered a vehicle and then checked the list of vehicles. We found the data of the vehicle we registered.

Table 4: Test Case Sample

Test Case ID: TC-002 Test Type: Unit Testing Component: NCP Vehicle Registration	Test Item: NCP Vehicle Registration Test Case Name: Test Case NCP Vehicle Registration Release Version: 1.0
Test Case Description: This module includes the registration process of NCP vehicles.	
Pre-Condition	Post Condition
User must be logged in	Data should be saved in the database
Input Data/Event	Expected output Data
Personal Details: Vehicle Details:	Data of this vehicle should be shown in the registered vehicles list
Test Case Result	
Data found	
Actual Input Data/Event	Actual Output Data/Event
Personal Details: Vehicle Details:	Data found
Test Case Result	
Test is performed successfully	

Here are the visuals of the test.

The screenshot shows a web application interface for the Excise and Taxation Department. On the left is a dark sidebar with a user profile for 'Munawar Johar (Excise User)' and a menu with options like 'General Settings', 'NCP Vehicles', 'Vehicle Registration', 'Challans', 'Owner Info', 'Vehicle', 'Vehicle Ownership', and 'Challan Detail'. The main content area is titled 'Excise and Taxation Department' and 'Developed by Students of CS Department, UOBS'. It features a 'Add Vehicle Registration' form with three steps: 1. Owner Info, 2. Vehicle Info, and 3. Vehicle Registration. The 'Owner Info' step is active, showing fields for CNIC (97823-6583112-7), Owner Name (John), Father's Name (Doe), Contact No (00000000000), Email (gmail@gmail.com), and Address (University of Baltistan Skardu). A blue 'Next »' button is at the bottom.

Figure 41: NCP Vehicle Registration

Excise and Taxation

Munawar Johar (Excise User)

Search

General Settings

NCP Vehicles

Vehicle Registration

Challans

Owner Info

Vehicle

Vehicle Ownership

Challan Detail

EXAMPLES

Calendar

Add Vehicle Registration

1 Owner Info 2 **Vehicle Info** 3 Vehicle Registration

Chasis No
54321

Engine No
12345

Color
Silver

Manufacturer
Toyota

Select Vehicle Type
Car

Select Vehicle Model
XLI

Vehicle CC:
1501 - 2000

« Previous Next »

Figure 42: NCP Vehicle Registration

Excise and Taxation

Munawar Johar (Excise User)

Search

General Settings

NCP Vehicles

Vehicle Registration

Challans

Owner Info

Vehicle

Vehicle Ownership

Challan Detail

EXAMPLES

Calendar

Add Vehicle Registration

1 Owner Info 2 Vehicle Info 3 **Vehicle Registration**

Available Reg#:
SKD202

Purchase Date
11-07-2024

Registration Date
11-07-2024

Reg Expiry Date
11-07-2025

Registration Status?
☒

« Previous

Submit

Figure 43: NCP Vehicle Registration

After clicking on submit, data will be saved in database.

The screenshot displays the 'Excise and Taxation Department' web application. The left sidebar shows the user 'Munawar Johar (Excise User)' and a menu with options like 'General Settings', 'NCP Vehicles' (selected), 'Vehicle Registration', 'Challans', 'Owner Info', 'Vehicle', 'Vehicle Ownership', and 'Challan Detail'. The main content area has a header 'Excise and Taxation Department' and a subtitle 'Developed by Students of CS Department, UOBS'. Below this is an 'Add Owner' button and a 'List of Owners' section. The 'List of Owners' section includes a search bar and a table with columns: S.No, Name, Father's Name, CNIC, Cell#, Email, and Address. The table contains two entries: one for 'John Doe' and another for 'Naveed Siddique'.

S.No	Name	Father's Name	CNIC	Cell#	Email	Address
1	John	Doe	97823-6583112-7	00000000000	gmail@gmail.com	University of Baltistan Skardu
2	Naveed Siddique	Muhammad Siddique Jan	71103-2133145-3	03334523256	naveed.siddique@hotmail.com	Naveed, House, Lahore

Figure 44: NCP Vehicle Registration

We can see that the recent entry is present in the list now.

5.3. Installation Process Description

Our system is very easy to use in terms of user experience and installation process. First of all, a reliable server is needed for the placement of data. After that they are set to go with the web-based system. There is no need to install heavy applications to run the system. However, a mobile application has to be installed, in order to retrieve updates from the server. The interface of both website and mobile application are as follows:

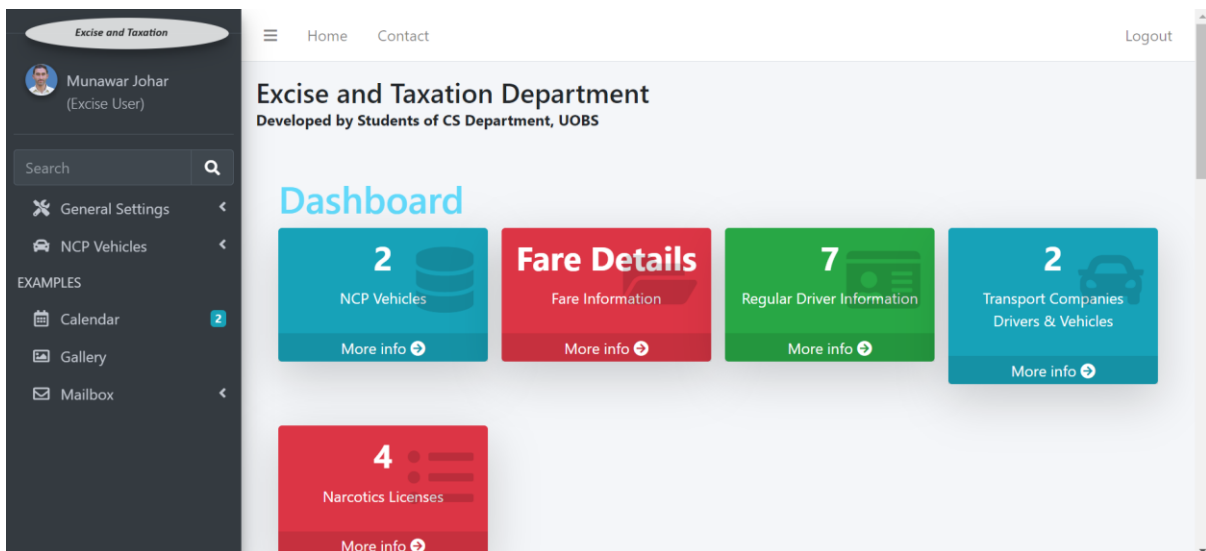


Figure 45: Website Homepage

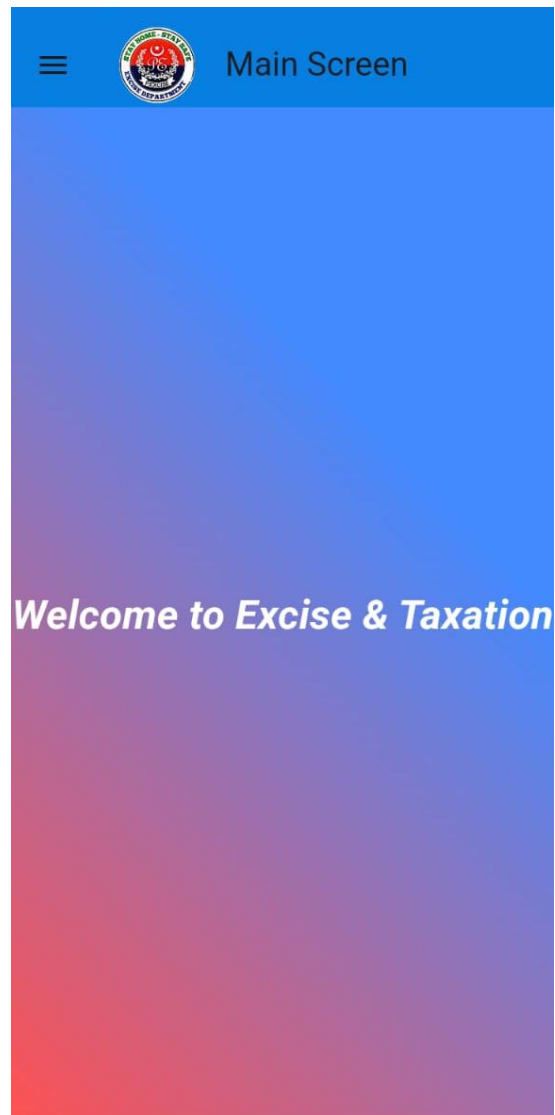


Figure 46: Mobile Application Homepage

Chapter 6

Conclusion and Future Work

6.1. Evaluation of Objectives and Aims

Our project successfully achieved several objectives outlined in the initial proposal, including centralizing data management, automating processes, and enhancing transparency. We increased data accuracy and optimized processes by putting database management tools and automated renewal / notification systems into place. Nevertheless, obstacles like resource limitations and system integration made it difficult to fully achieve some goals. In order to further refine the system and better suit the department's changing demands, it will be imperative that these difficulties are addressed going forward.

6.2. Evaluation of Project Management

Throughout the development of the Excise and Taxation Management System for Baltistan, effective project management played a crucial role in ensuring the successful completion of the project. We are lucky that we were under supervision of one of the most supportive supervisors. He divided the responsibilities according to our capabilities on day one. This allocation of responsibilities involved close coordination between the project supervisor and our group member, ensuring alignment with project objectives and timely delivery of milestones.

Project Coordination:

Regular zoom meetings were scheduled with the project supervisor to discuss project progress, address issues or concerns, and receive guidance on project direction. These meetings provided valuable opportunities to receive feedback, clarify requirements, and make necessary adjustments to the project plan. Additionally, communication channels such as WhatsApp, text messaging and zoom meetings were utilized to maintain continuous communication and collaboration among group.

Task Allocation and Monitoring:

Tasks were effectively allocated among project partners based on individual expertise and capabilities. The project supervisor played a pivotal role in overseeing task assignments, ensuring that each partner had a clear understanding of their responsibilities and deadlines. Regular progress updates were shared among team members to monitor task completion

and identify any bottlenecks or delays. Any deviations from the project plan were promptly addressed through proactive problem-solving and resource reallocation.

Close Collaboration:

In addition to the structured project management approach, a significant contributing factor to the project's success was the close collaboration among team members. Three out of the four project members resided in a hostel, enabling frequent face-to-face meetings and collaborative work sessions. This physical proximity facilitated real-time problem-solving, brainstorming sessions, and efficient decision-making, ultimately accelerating the project's progress. We constantly pondered the project while sitting, sleeping, waking up, and even during meals and drinks. The ability to work closely together fostered a strong sense of frankness and cohesion within the team, enhancing communication, trust, and productivity. Through this collaborative environment, project members were able to leverage each other's strengths, support one another through challenges, and collectively drive the project towards its objectives.

A Regret:

While we diligently kept in touch with the assistant, regrettably, we failed to engage directly with the Director of the Excise and Taxation Department. This lapse in communication led to a misalignment of expectations and priorities between the Director and his assistant. As a consequence, we encountered discrepancies in project requirements and objectives, as each stakeholder possessed different visions for the project. The Director's perspective, which was not adequately addressed due to our oversight, presented alternative insights and preferences that were crucial for aligning our project deliverables with the high goals of the department. Moving forward, we recognize the importance of establishing direct communication channels with all key stakeholders, ensuring that their perspectives are heard, and their expectations are adequately addressed to foster a unified vision for project success.

Another regret is that, we wasted months coding in plain PHP. But after winter vacations our supervisor suggested to use locally built code architecture. If we did the same in start we would have finished the project.

Overall, effective project management, characterized by close coordination, task allocation, and collaborative decision-making facilitated the successful progress of the Excise and Taxation Management System for Baltistan. The collaborative efforts of the project supervisor and group members ensured that project objectives were met and stakeholders' expectations were addressed. We hope that the rest of the project will be completed on time with the help of our project supervisor.

6.3. Discussion

Several of the original proposal's goals, such as centralizing data management, automating procedures, and improving transparency, were successfully met by our project. But we made a mistake. While we diligently kept in touch with the assistant, regrettably, we failed to engage directly with the Director of the Excise and Taxation Department. This lapse in communication led to a misalignment of expectations and priorities between the Director and his assistant. As a consequence, we encountered discrepancies in project requirements and objectives, as each stakeholder possessed different visions for the project. The Director's perspective, which was not adequately addressed due to our oversight, presented alternative insights and preferences that were crucial for aligning our project deliverables with the high goals of the department. Moving forward, we recognize the importance of establishing direct communication channels with all key stakeholders, ensuring that their perspectives are heard, and their expectations are adequately addressed to foster a unified vision for project success.

6.4. Future Work

Besides all the progress in the project we understand our capabilities with limited exposure and no industry experience. We may have built a robust system but there is always the possibility of betterment. There are many areas which might need improvement. Some of them are being discussed here.

6.4.1. UI / UX:

We tried our best to provide a smooth and responsive system, but, as none of us has industry experience so there might be some flaws in it. It can only be tested by the end users. So, after using this system for a while, with the end user's feedback, user experience can be improved.

6.4.2. Availability for Public:

For now, the system can be accessed by the department only. But in future, they will make it available for public as some modules are specially made for public. For example, they will be able to register a complaint or they will be able to verify their driving license from the comfort of their homes.

Moreover, there is also a possibility of allowing the public to apply for their registration online. This will be a great initiative towards a luxurious future.

6.4.3. Module/s for Future:

In one of our first meetings, they have told us to develop the driver's license verification module and regular vehicle module. But they are not willing these modules to make it live right now. They are saying to make it happen later.

Chapter 7

References

- [1] <https://excise.punjab.gov.pk/>
- [2] <https://www.kpexcise.gov.pk/app/>
- [3] <https://excise.gos.pk/>
- [4] <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- [5] <https://developer.mozilla.org/en-US/docs/Web/CSS>
- [6] <https://www.php.net/docs.php>
- [7] <https://api.jquery.com/>
- [8] <https://getbootstrap.com/docs/4.1/getting-started/introduction/>
- [9] <https://code.visualstudio.com/docs>
- [10] <https://dev.mysql.com/doc/>

Chapter 8

Appendix

8.1. A Guide to Excise & Taxation Management System

8.1.1. How to Login?

Login is the first step in this system. The user will have to enter his credential and click on the “Sign In” Button.

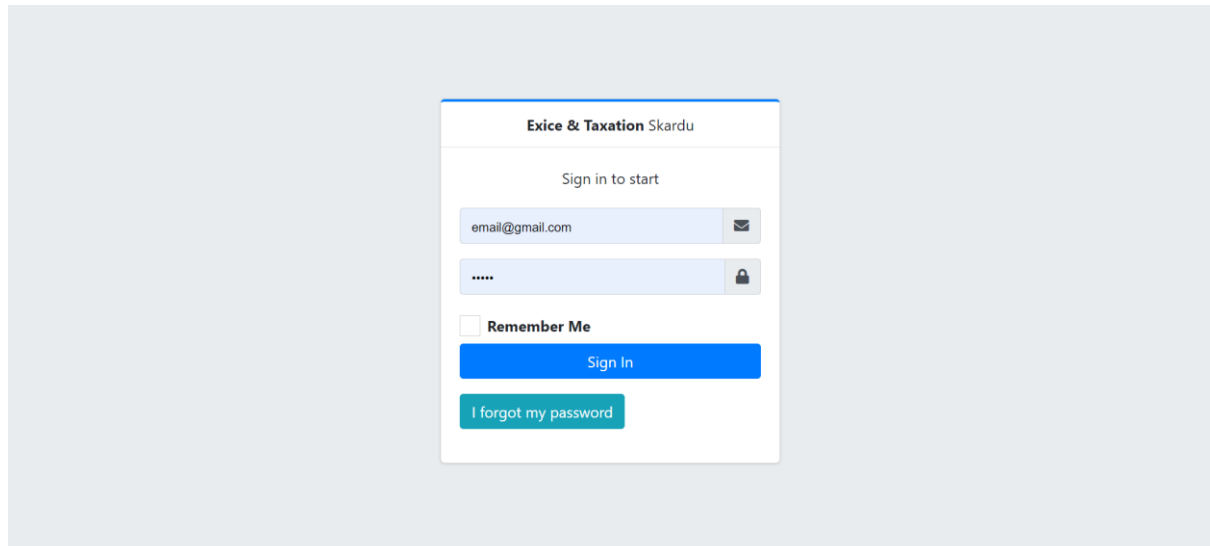


Figure 47: Login View

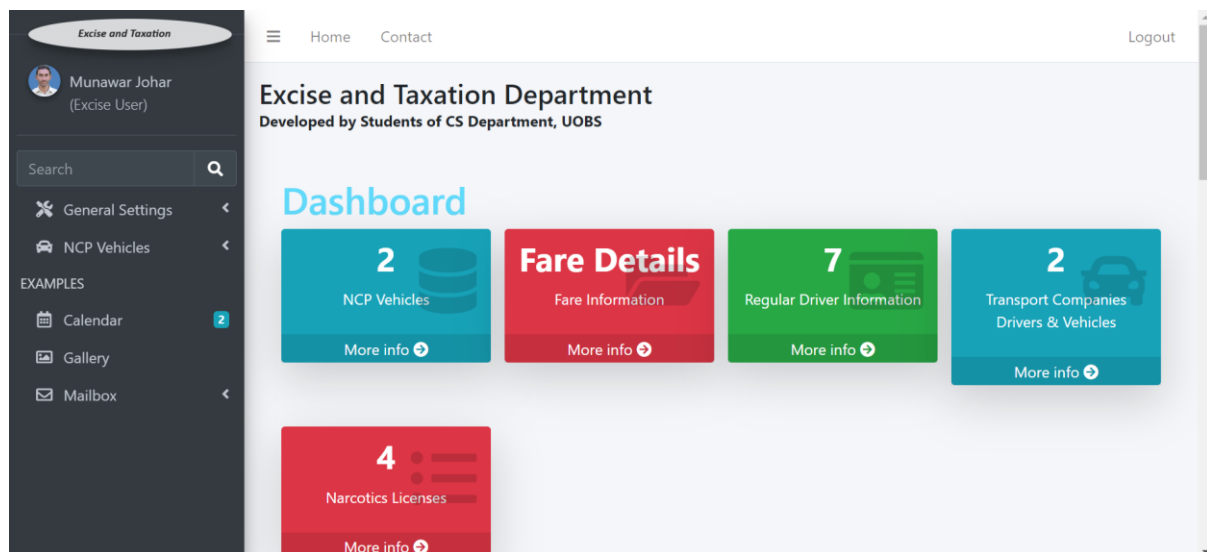


Figure 48: Login Successful (Homepage)

8.1.2. Accessing NCP Vehicles Module

First of all, the user has to login. After login, the homepage will be appeared as shown in Figure 48.

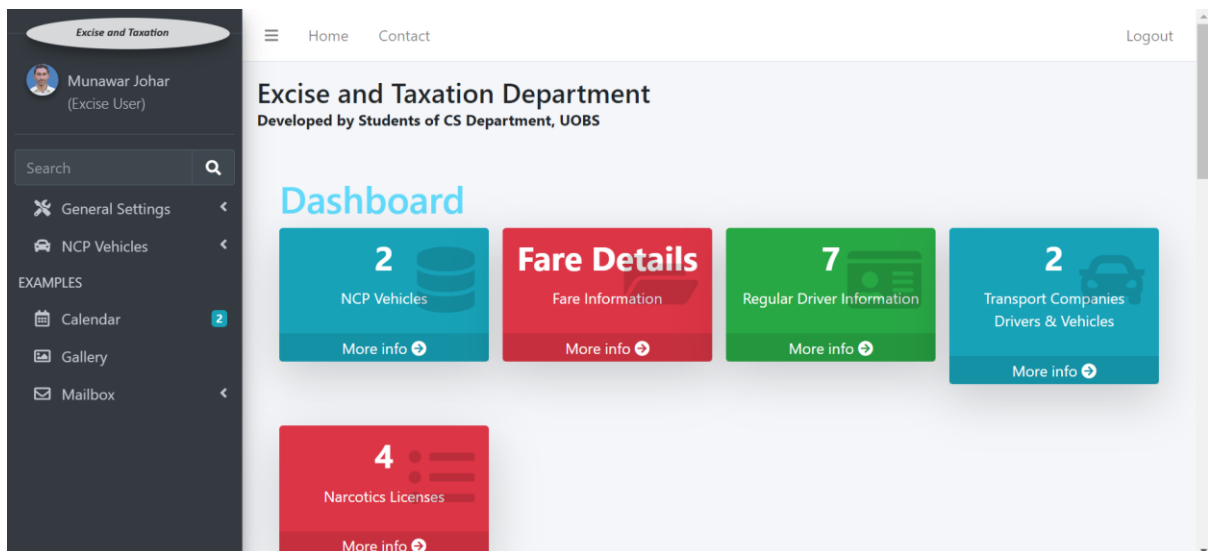


Figure 49: Homepage

After reaching homepage, user has to select his desired module. Let's say he clicked on NCP Vehicles. Figure 50 will be the result then.

Figure 50: Vehicle Registration Form

After filling the form, click next and then fill all required information and then submit it. It is as simple as this.

Excise and Taxation

Munawar Johar
(Excise User)

Search

General Settings

NCP Vehicles

Vehicle Registration

Challans

Owner Info

Vehicle

Vehicle Ownership

Challan Detail

EXAMPLES

Calendar

Add Vehicle Registration

1 Owner Info 2 **Vehicle Info** 3 Vehicle Registration

Chasis No
54321

Engine No
12345

Color
Silver

Manufacturer
Toyota

Select Vehicle Type
Car

Select Vehicle Model
XLI

Vehicle CC:
1501 - 2000

« Previous Next »

Figure 51: Vehicle Registration Form

Excise and Taxation

Munawar Johar
(Excise User)

Search

General Settings

NCP Vehicles

Vehicle Registration

Challans

Owner Info

Vehicle

Vehicle Ownership

Challan Detail

EXAMPLES

Calendar

Add Vehicle Registration

1 Owner Info 2 Vehicle Info 3 **Vehicle Registration**

Available Reg#:
SKD202

Purchase Date
11-07-2024

Registration Date
11-07-2024

Reg Expiry Date
11-07-2025

Registration Status?
☒

« Previous

Submit

Figure 52: Vehicle Registration Form

After clicking on submit, data will be saved in database.

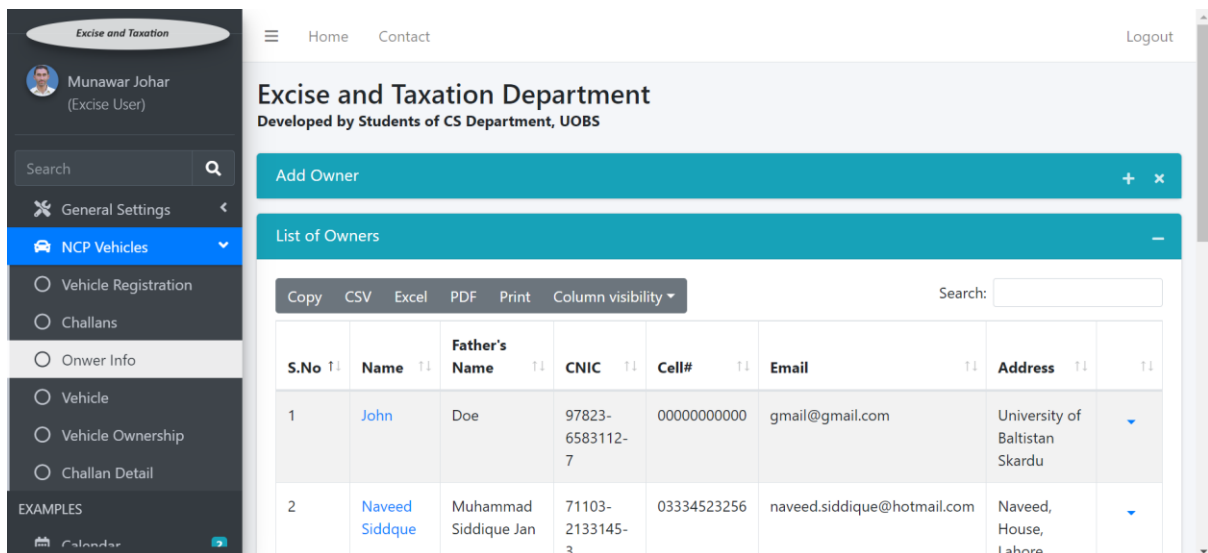


Figure 53: Entry Added in Database Successfully

The other modules can also be operated like this. We saw from above guide (section 8.1) that the system is very easy to navigate.

8.2. Code Snippets

```

expire.php  expire_soon.php  verify_license.php  drivers.php X
> xampp > htdocs > Excise > ncp_vehicles > drivers > drivers.php
90 <div class="card card-info">
91 <div class="card-header">
92 <h3 class="card-title">Driver Licenses</h3>
93 <div class="card-tools">
94 <button type="button" class="btn btn-tool" data-card-widget="collapse" title="Collapse">
95 <i class="fas fa-minus"></i>
96 </button>
97 </div>
98 </div>
99 <!-- /.card-header -->
100 <div class="card-body">
101 <table id="example1" class="table table-bordered table-striped">
102 <thead>
103 <tr>
104 <th>S.No</th>
105 <th>Name</th>
106 <th>Issue Date</th>
107 <th>Date of Birth</th>
108 <th>CNIC</th>
109 <th>Father's Name</th>
110 <th>License Number</th>
111 <th>Type</th>
112 <th>Expiry Date</th>
113 <th>Edit</th>
114 </tr>
115 </thead>
116 <tbody>
117 <?php
118 $sno = 0;
119 $drivers = getMultipleRows("drivers", "*", "1 ORDER BY id DESC");
120 foreach($drivers as $driver)
121 {
122     $sno++;
123 }

```

Figure 54: Code snippet of driver's verification module

```

license.php x drivers.php fare_detail.php calculate_fare.php calculate_freight.php fare_complaints.php vehicle_ownership.php challan.php x
C: > xampp > htdocs > Excise > ncp_vehicles > vehicles > challan.php
241
242
243 <div class="card card-info">
244 <div class="card-header">
245 <h3 class="card-title">List of Challans</h3>
246 <div class="card-tools">
247 <button type="button" class="btn btn-tool" data-card-widget="collapse" title="Collapse">
248 <i class="fas fa-minus"></i>
249 </button>
250 </div>
251 </div>
252 <!-- /.card-header -->
253 <div class="card-body">
254 <table id="example1" class="table table-bordered table-striped">
255 <thead>
256 <tr>
257 <th>S.No</th>
258 <th>Challan No</th>
259 <th>Total Amount</th>
260 <th>Challan Date</th>
261 <th>Pay Date</th>
262 <th>Payment Staus</th>
263 <th>Narration</th>
264 <th></th>
265 </tr>
266 </thead>
267
268 <tbody>
269 <?php
270 $sno = 0;
271
272 // $members = getMultipleRows("`members` f
273 // join users u on u.id = f.user_id"
274 // , "f.*, u.email,u.active,u.picture");
275 $challans = getMultipleRows("`challan`","*", "1 order by id desc");

```

Figure 55: Code snippet of challan / TR6

```

EXPLORER Welcome vehicle_type.php vehicle.php drivers.php verify_license.php vehicle_ownership.php fare_complaints.php x
v EXCISE ncp_vehicles > fare > fare_complaints.php
> home 121 <div class= card card-info <?php echo $collapse > >
> images 201 </div>
> includes 202
> ncp_vehicles 203 <div class="card card-info">
> drivers 204 <div class="card-header">
> drivers.php 205 <h3 class="card-title">Complaints</h3>
> expire_soon.php 206 <div class="card-tools">
> expire.php 207 <button type="button" class="btn btn-tool" data-card-widget="collapse" title="Collapse">
> vehicle_owners... 208 <i class="fas fa-minus"></i>
> verify_license.p... 209 </button>
210 </div>
211 </div>
212 <div class="card-body">
213 <table id="example1" class="table table-bordered table-striped">
214 <thead>
215 <tr>
216 <th>S.No</th>
217 <th>Passenger Name</th>
218 <th>CNIC</th>
219 <th>Email</th>
220 <th>Phone Number</th>
221 <th>Complaint Type</th>
222 <th>Complaint Details</th>
223 <th>Status</th>
224 <th>Date Filed</th>
225 <th>Date Resolved</th>
226 <th>Edit</th>
227 </tr>
228 </thead>
229 <tbody>
230 <?php
231 $sno = 0;
232 $complaints = getMultipleRows("`fare_complaints`","*", "1 ORDER BY complaint_id DESC");
233 foreach($complaints as $complaint)

```

Figure 56: Code Snippet of Fare Details module

```

<a class="btn btn-warning dropdown-toggle dropdown-icon" data-toggle="dropdown" href="#">
    <span class="caret">Action</span>
</a>
<div class="dropdown-menu text-center">

    <form action="" method="POST">
        <input type="hidden" name="registration_id" value="<?php echo $registration_id ?>">
        <input type="hidden" name="vehicle_id" value="<?php echo $vehicle_id ?>">
        <input type="hidden" name="owner_id" value="<?php echo $owner_id ?>">
        <input type="hidden" name="reg_no" value="<?php echo $vehicle_pool['reg_no'] ?>">
        <input type="hidden" name="challan_no" value="<?php echo $vehicle_challan['challan_no'] ?>">
        <input type="hidden" name="amount" value="<?php echo $vehicle_challan['total_amount'] ?>">
        <button type="submit" name="cch_vehicle_registration_edit" class="btn btn-info"><i
            class="far fa-edit"></i> &nbsp; Print Challan &nbsp; </button>
        <div class="dropdown-divider"></div>
        <button type="submit" onclick="return confirm('Do you want to delete this record?')"
            name="cch_vehicle_registration_delete" class="btn btn-danger"><i class="fas fa-trash"></i>
            Delete</button>
        <div class="dropdown-divider"></div>
        <button type="submit" name="cch_vehicle_registration" class="btn btn-success"><i
            class="far fa-edit"></i> &nbsp; Detail &nbsp; </button>
        <div class="dropdown-divider"></div>
    </form>

```

Figure 57: Code snippet of vehicle registration module

```

if(isset($_POST['cch_vehicle_ownership_edit']))
{
    $ownership_id = $_POST['ownership_id'];
    $edit_ownership = getSingleRow("vehicle_owner","*", "id = '$ownership_id'");
    $vehicle_id = $_POST['vehicle_id'];
    $owner_id = $_POST['owner_id'];

    $collapsed = "";
    $add = "Edit ";
}
else{
    $ownership_id="";
}

if(isset($_POST['cch_vehicle_owner_delete']))
{
    $ownership_id = $_POST['ownership_id'];
    $ownership_where = "is_current = 1 and id = '$ownership_id'";
    $sel_ownership_rows = no_of_rows("vehicle_owner" ,"*", $ownership_where);
    if($sel_ownership_rows ==0)
    {
        $table = "vehicle_owner";
        $where = "id = '$ownership_id'";

        if(delete($table, $where))
        {
            $msg = "Record Deleted";
            $alert = "success";
            $alert_icon = "check";
        }
    }
}

```

Figure 58: Code snippet of vehicle ownership class

```

if(isset($_POST['cch_challan_edit']))
{
    $challan_id = $_POST['challan_id'];
    $edit_challan = getSingleRow("challan","*", "id = '$challan_id'");
    $challan_date = date("d-m-Y", strtotime($edit_challan['challan_date']));
    $pay_date = date("d-m-Y", strtotime($edit_challan['pay_date']));
    $collapsed = "";
    $add = "Edit ";
}

if(isset($_POST['cch_challan_delete']))
{
    $challan_id = $_POST['challan_id'];
    $sel_challan_detail_rows = no_of_rows("challan_detail" ,"*", "challan_id = $challan_id");
    if($sel_challan_detail_rows ==0)
    {
        $table = "challan";
        $where = "id = '$challan_id'";

        if(delete($table, $where))
        {
            $msg = "Challan Deleted";
            $alert = "success";
            $alert_icon = "check";
        }
    }
}

```

Figure 59: Code snippet of Challan Class